

BFS Traversal

Step 1: Start at A

Legend

Complete

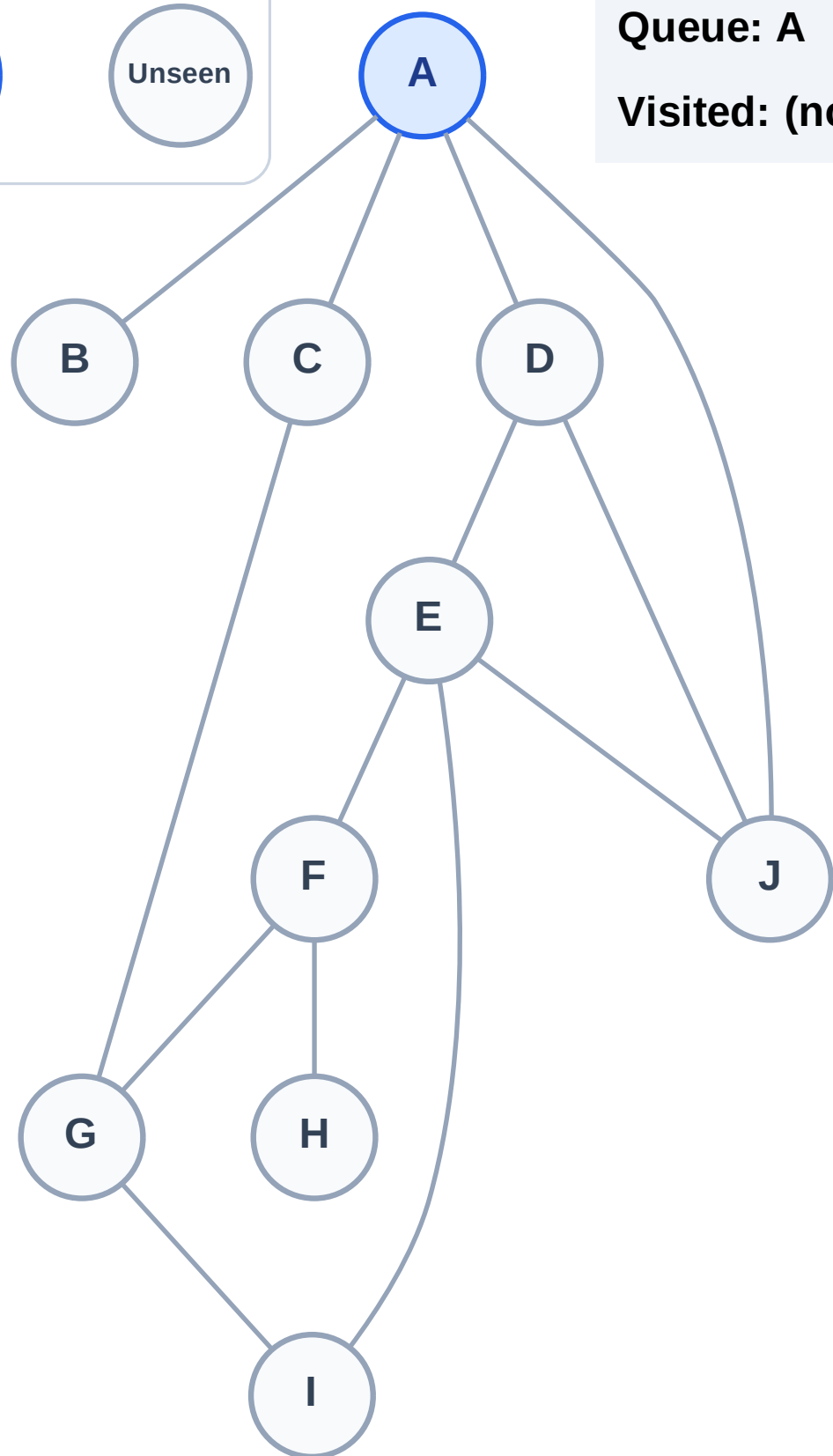
Processing

Discovered

Unseen

Queue: A

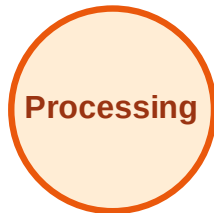
Visited: (none)



BFS Traversal

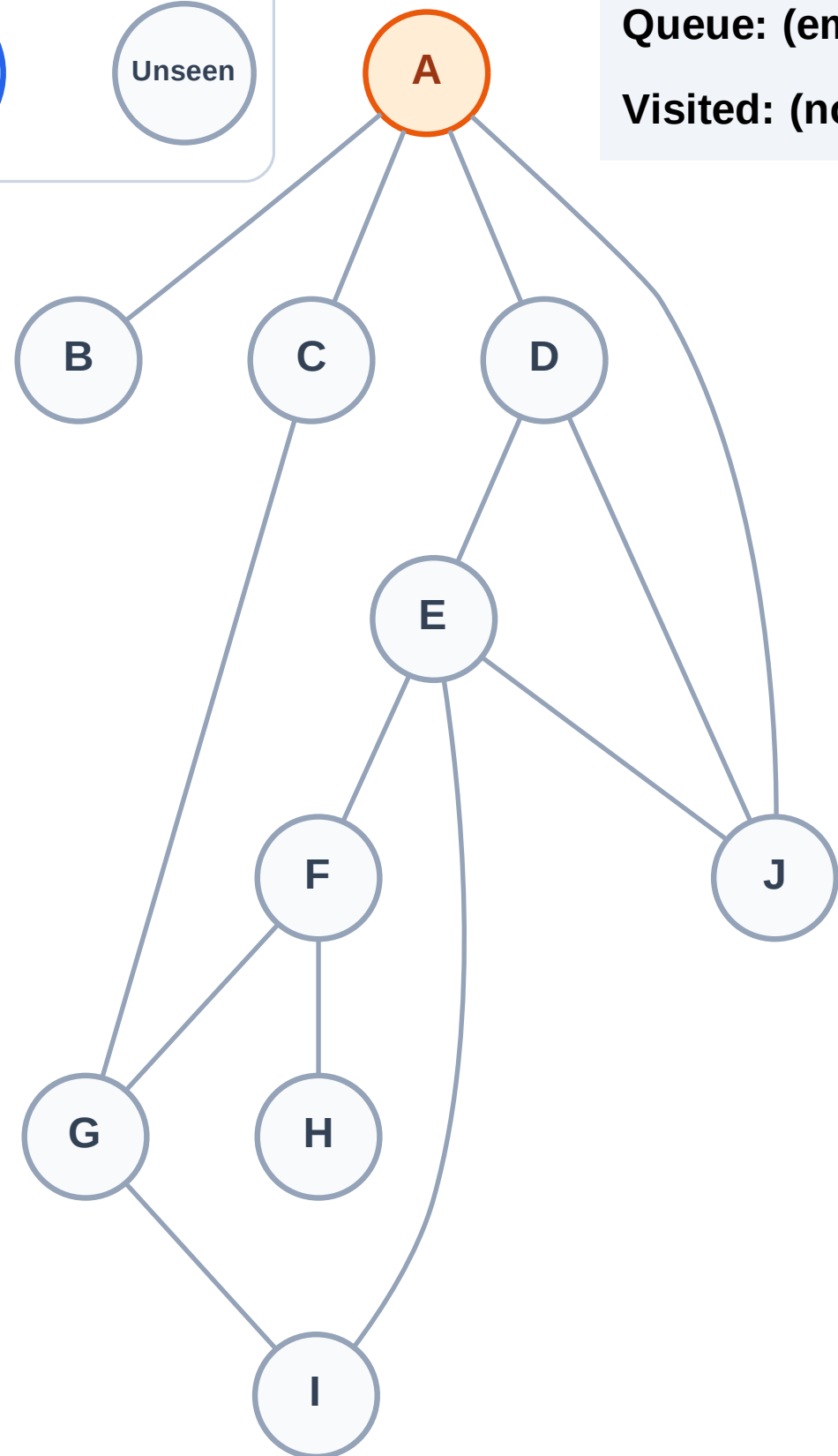
Step 2: Process node A

Legend



Queue: (empty)

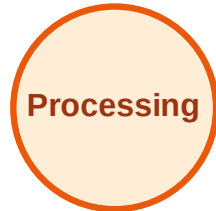
Visited: (none)



BFS Traversal

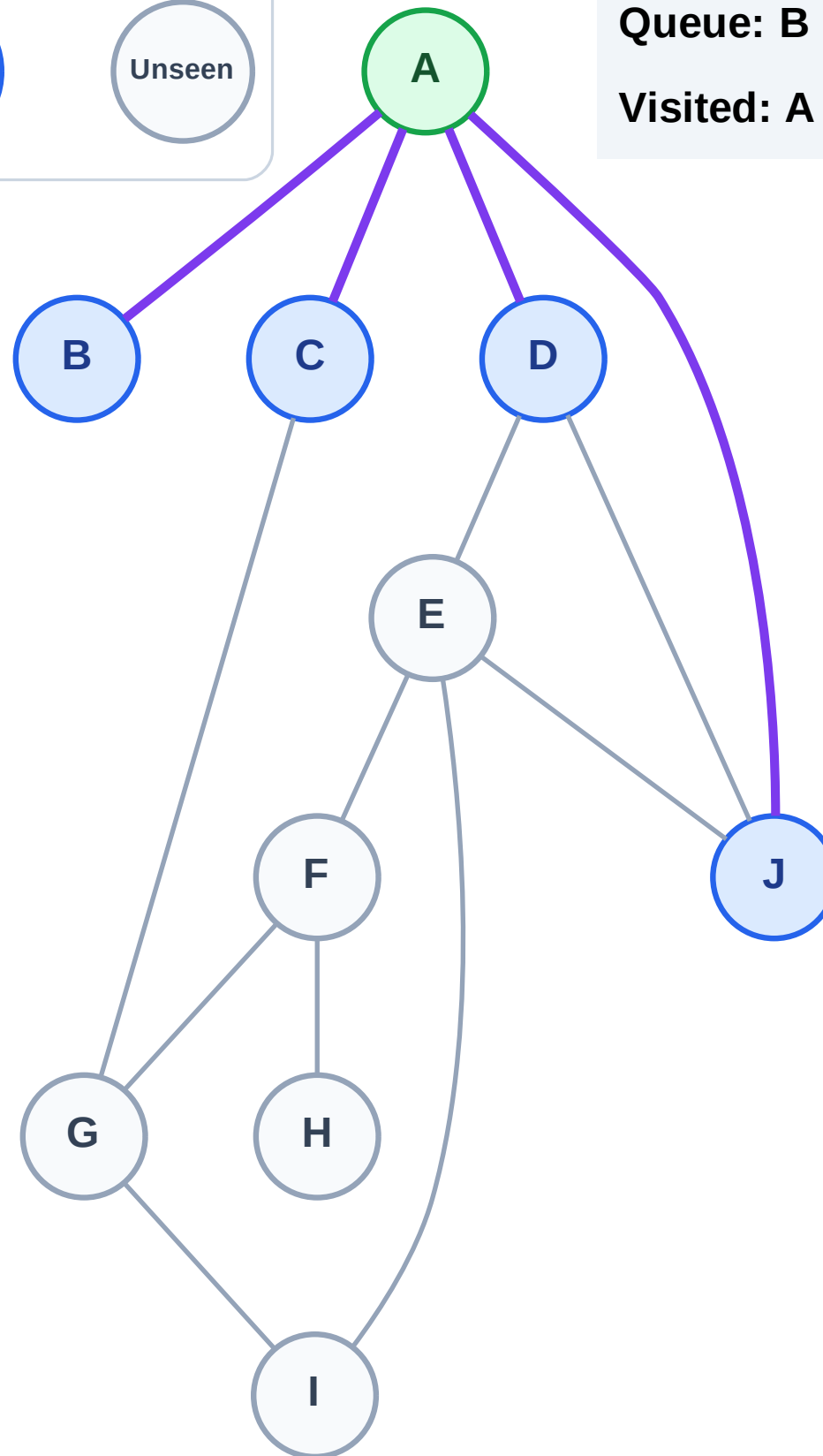
Step 3: Discover B, C, D, J from A

Legend



Queue: B → C → D → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

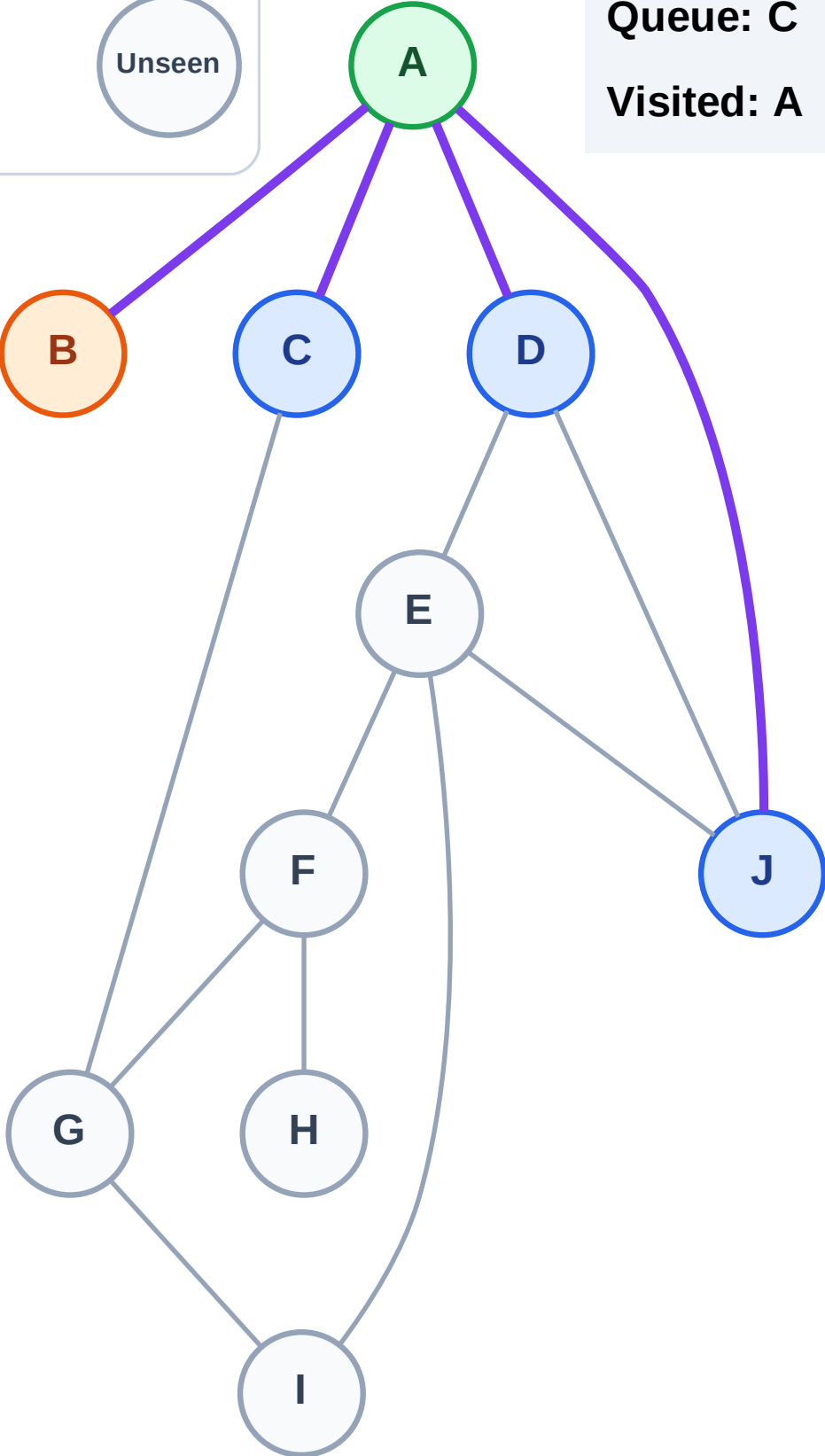
Processing

Discovered

Unseen

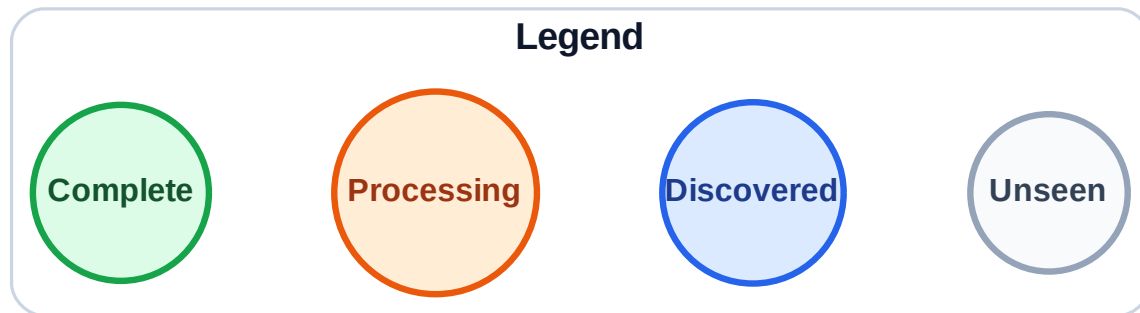
Queue: C → D → J

Visited: A

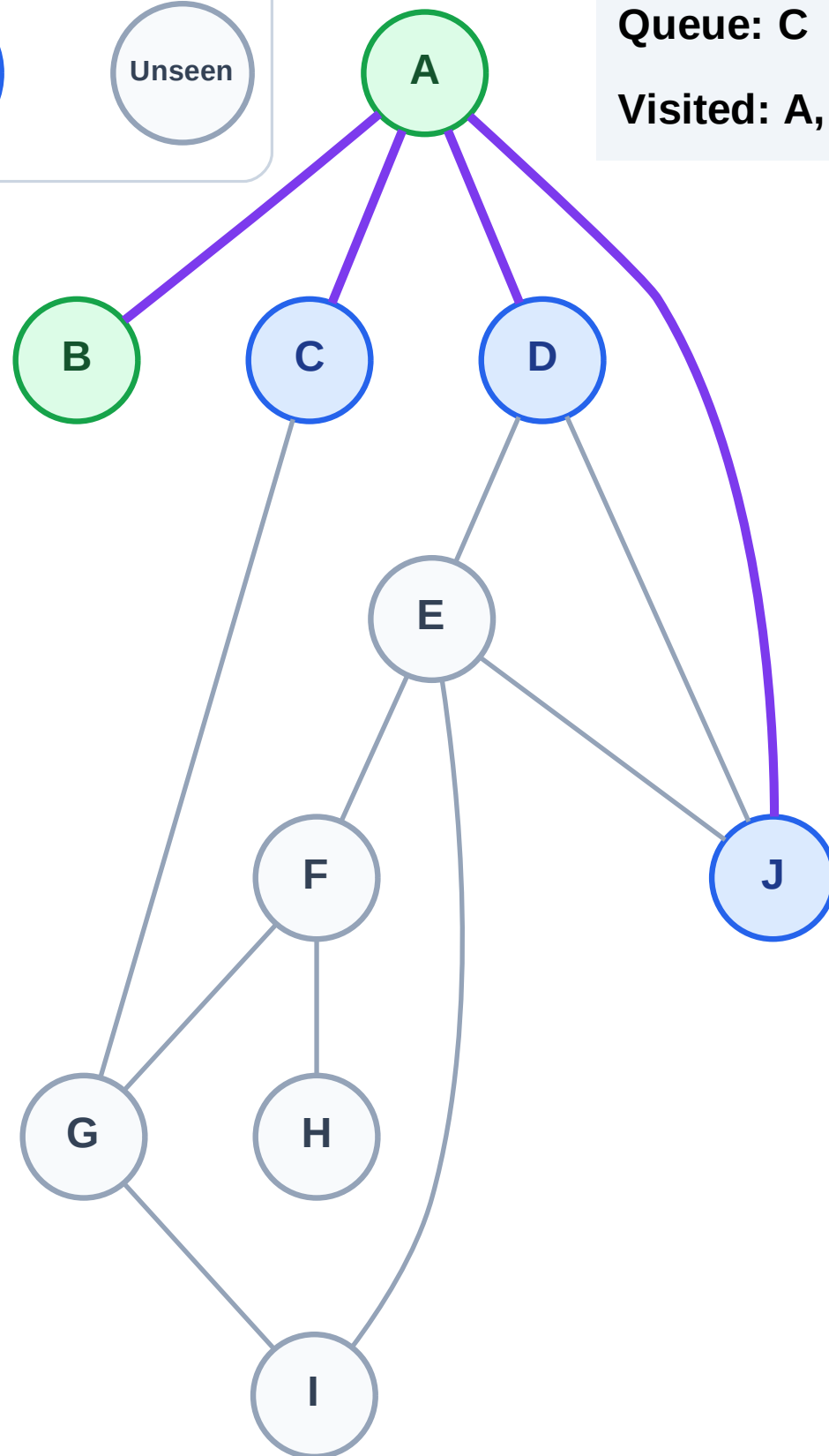


BFS Traversal

Step 5: No new neighbors from B

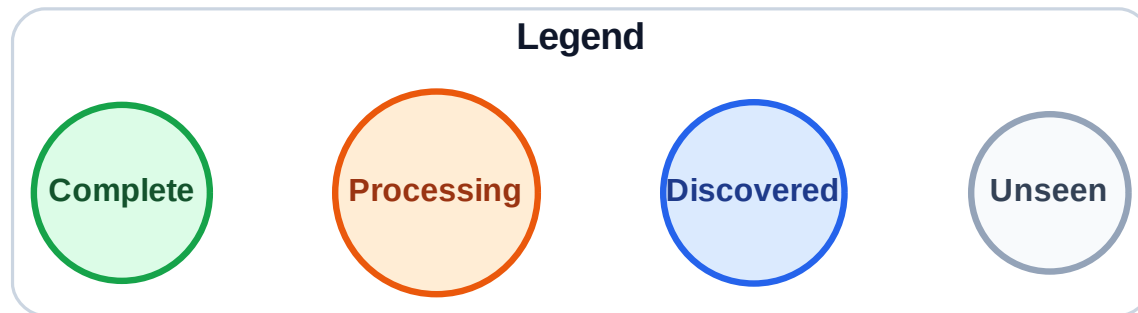


Queue: C → D → J
Visited: A, B



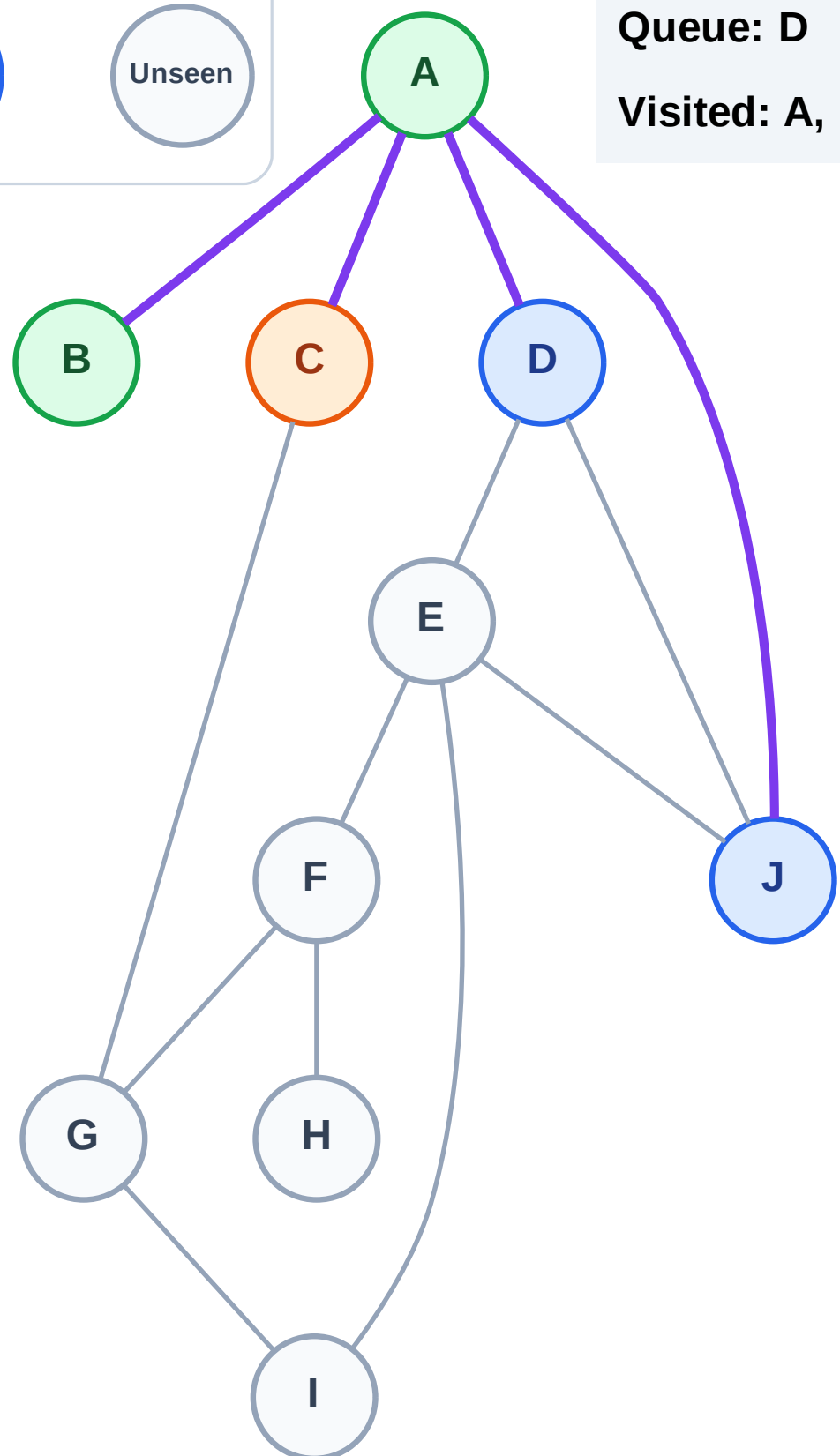
BFS Traversal

Step 6: Process node C



Queue: D → J

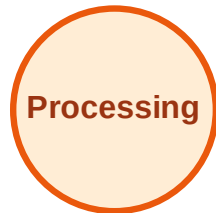
Visited: A, B



BFS Traversal

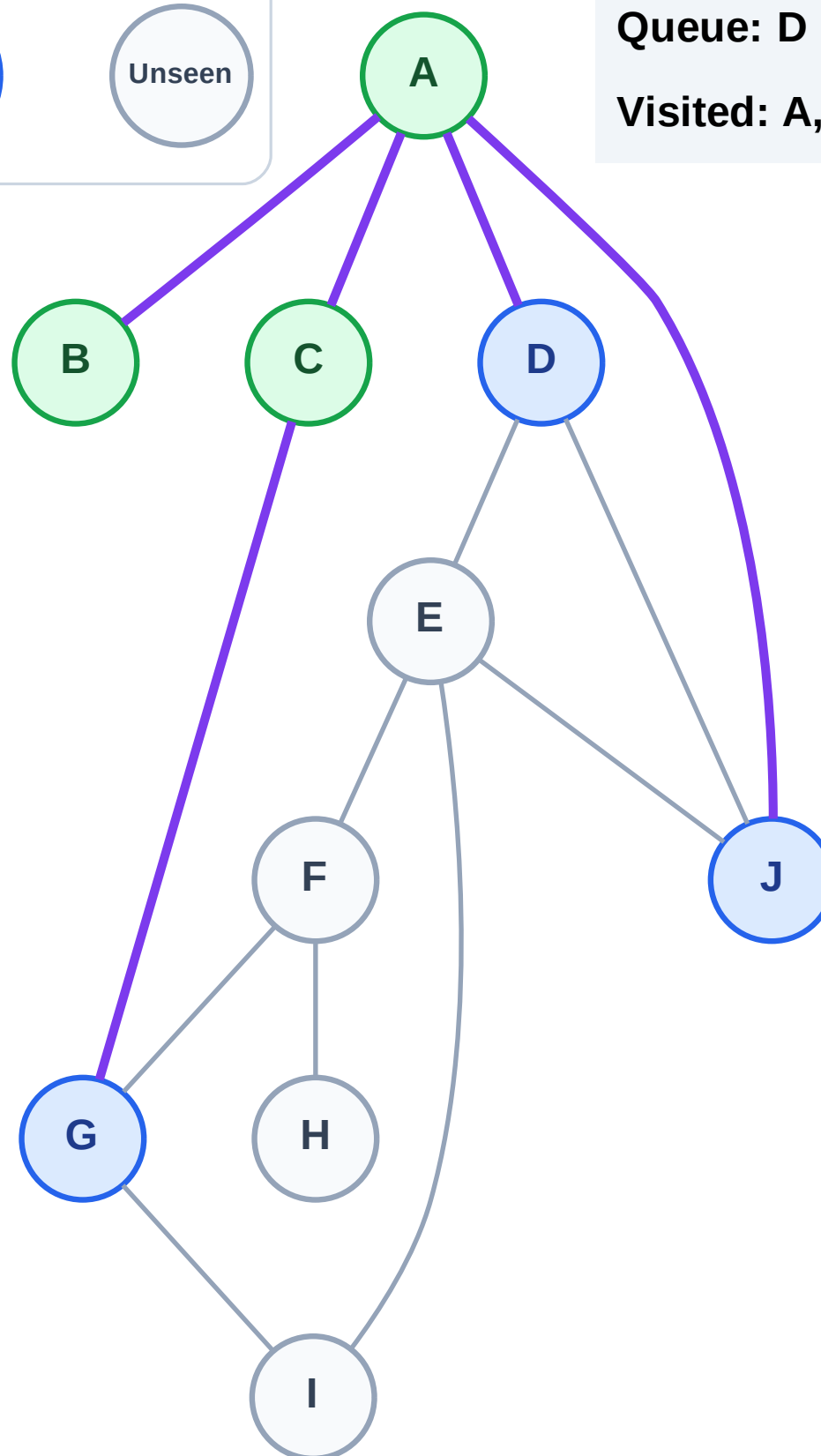
Step 7: Discover G from C

Legend



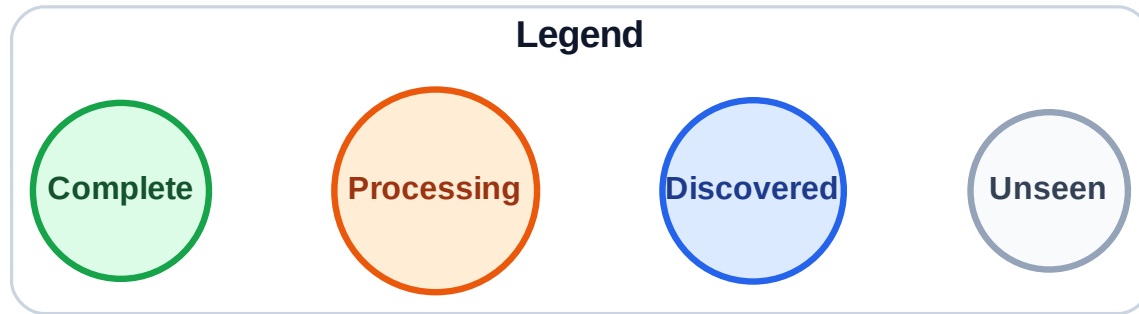
Queue: D → J → G

Visited: A, B, C



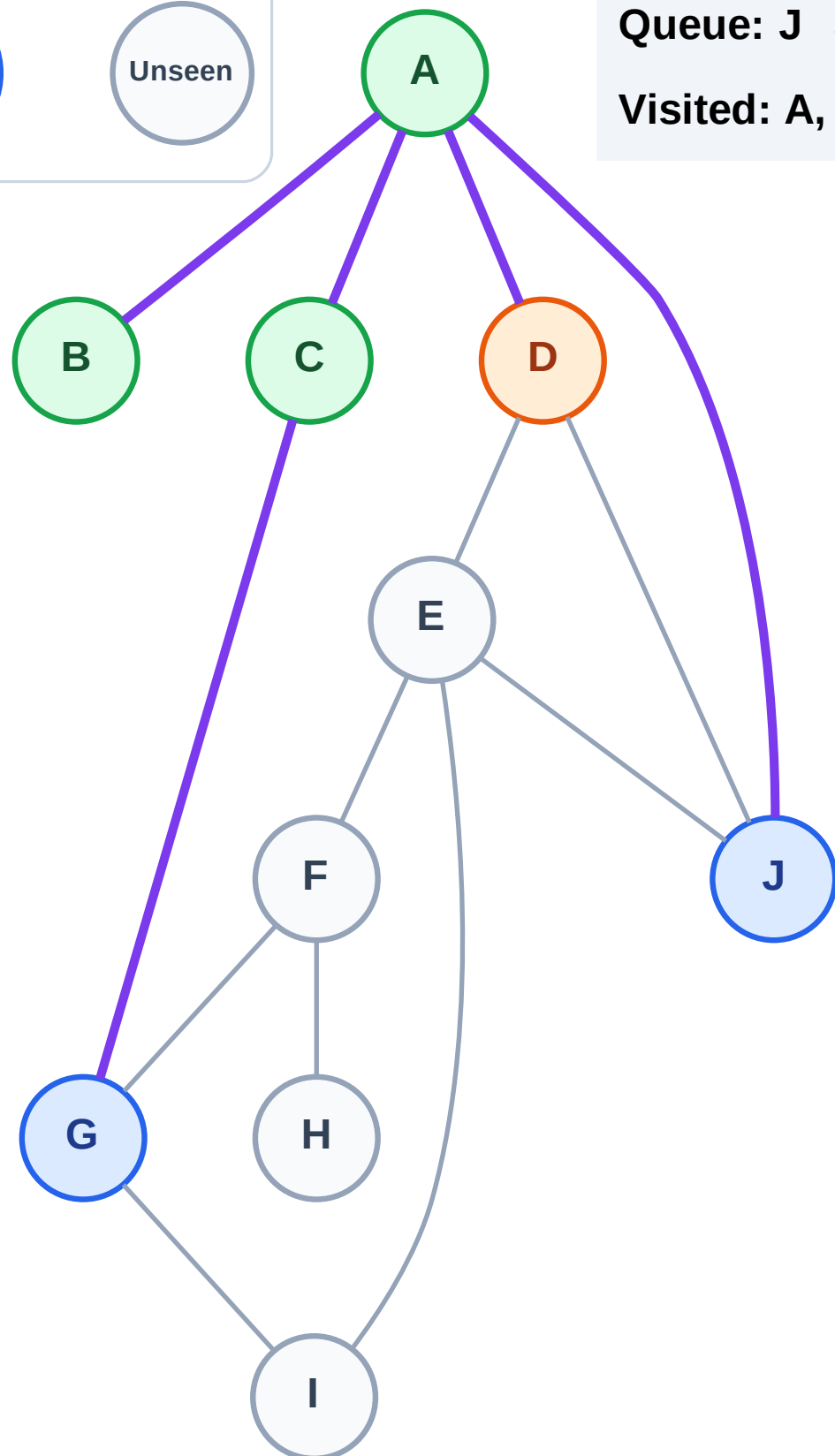
BFS Traversal

Step 8: Process node D



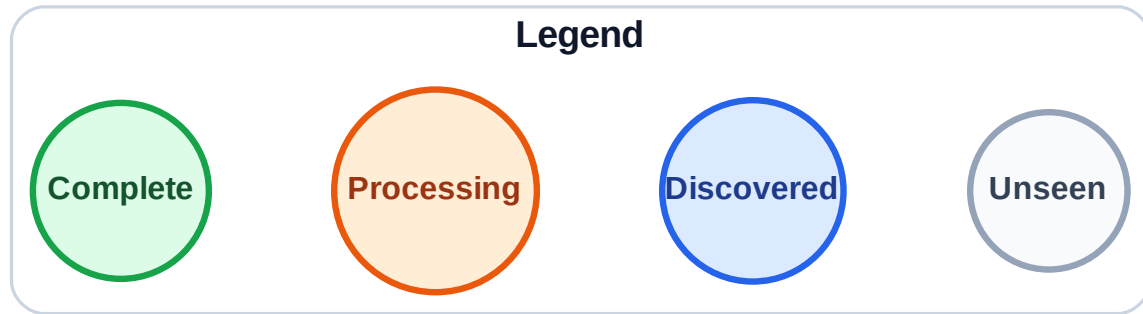
Queue: J → G

Visited: A, B, C

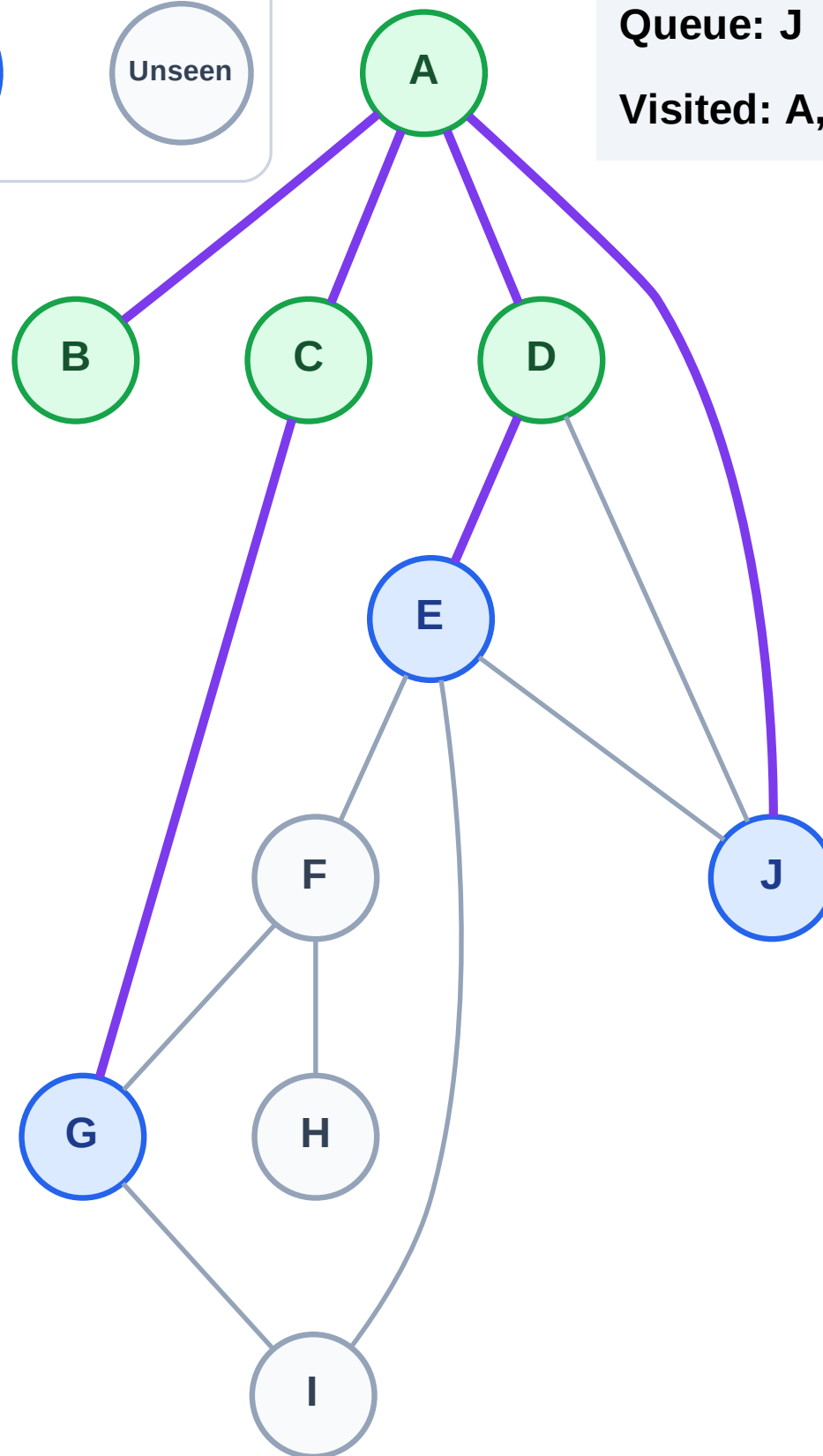


BFS Traversal

Step 9: Discover E from D



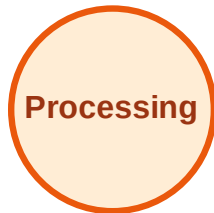
Queue: J → G → E
Visited: A, B, C, D



BFS Traversal

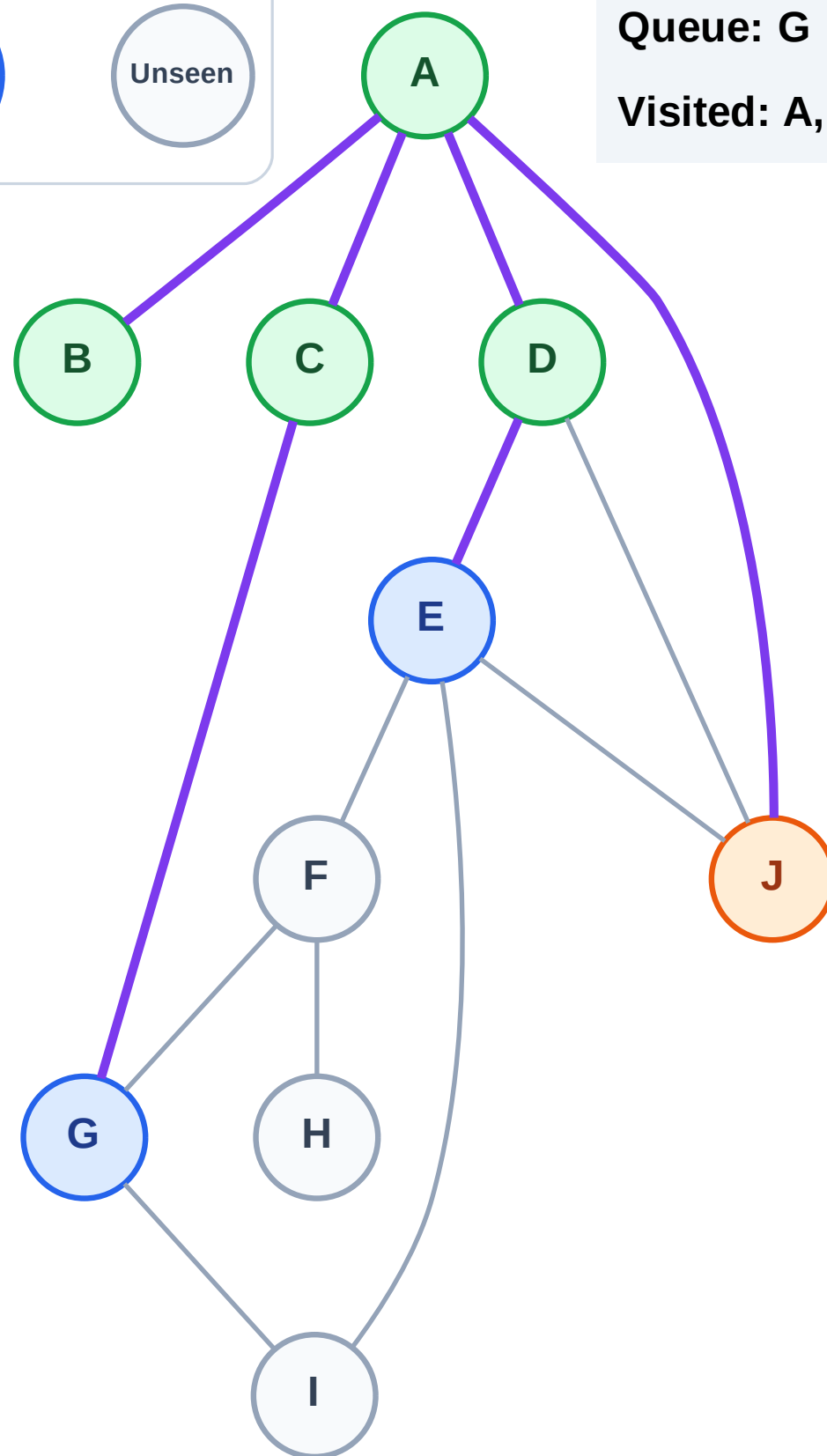
Step 10: Process node J

Legend



Queue: G → E

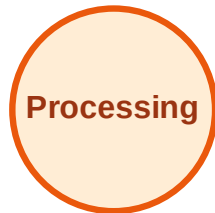
Visited: A, B, C, D



BFS Traversal

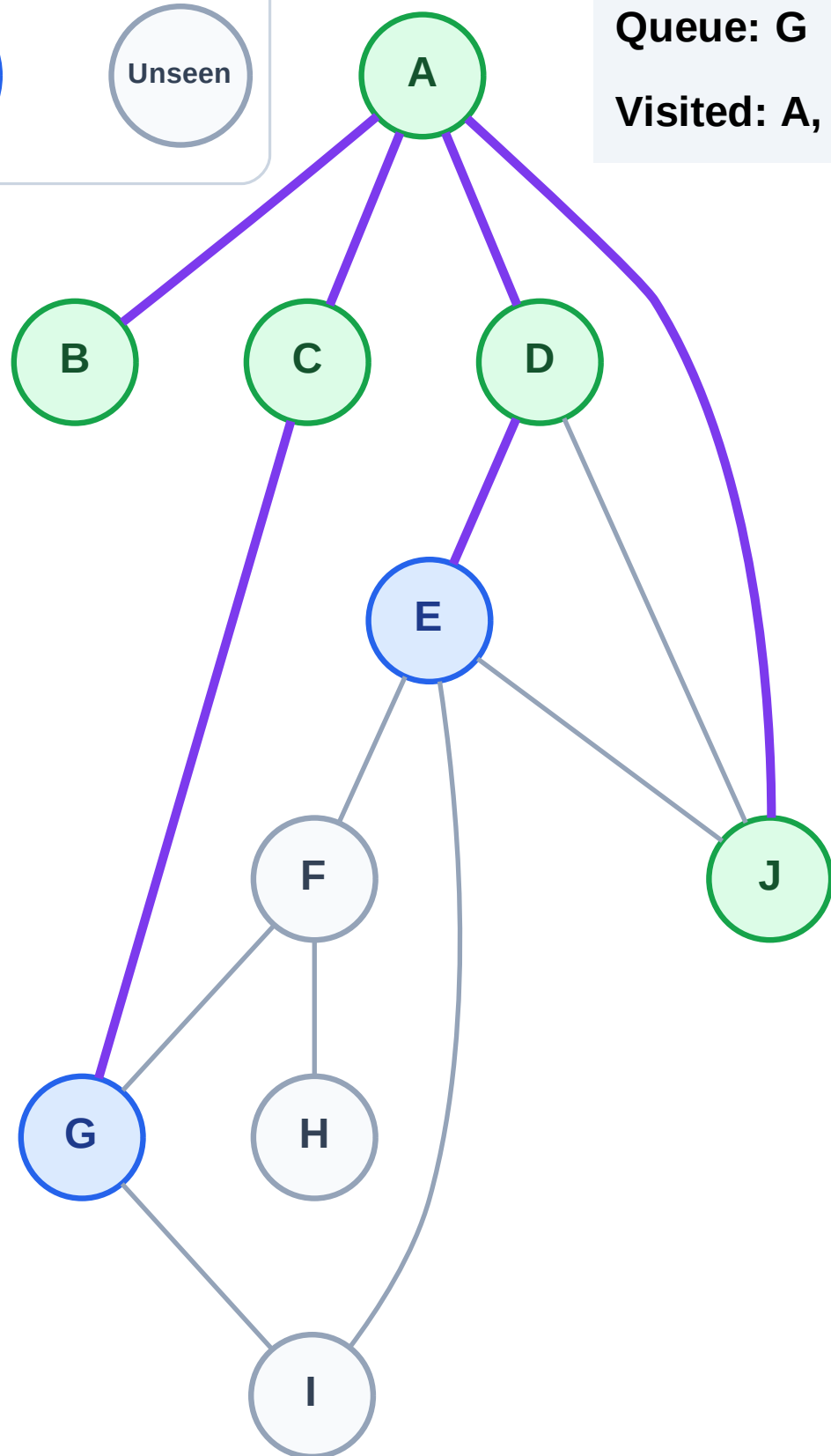
Step 11: No new neighbors from J

Legend



Queue: G → E

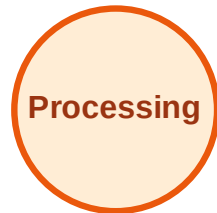
Visited: A, B, C, D, J



BFS Traversal

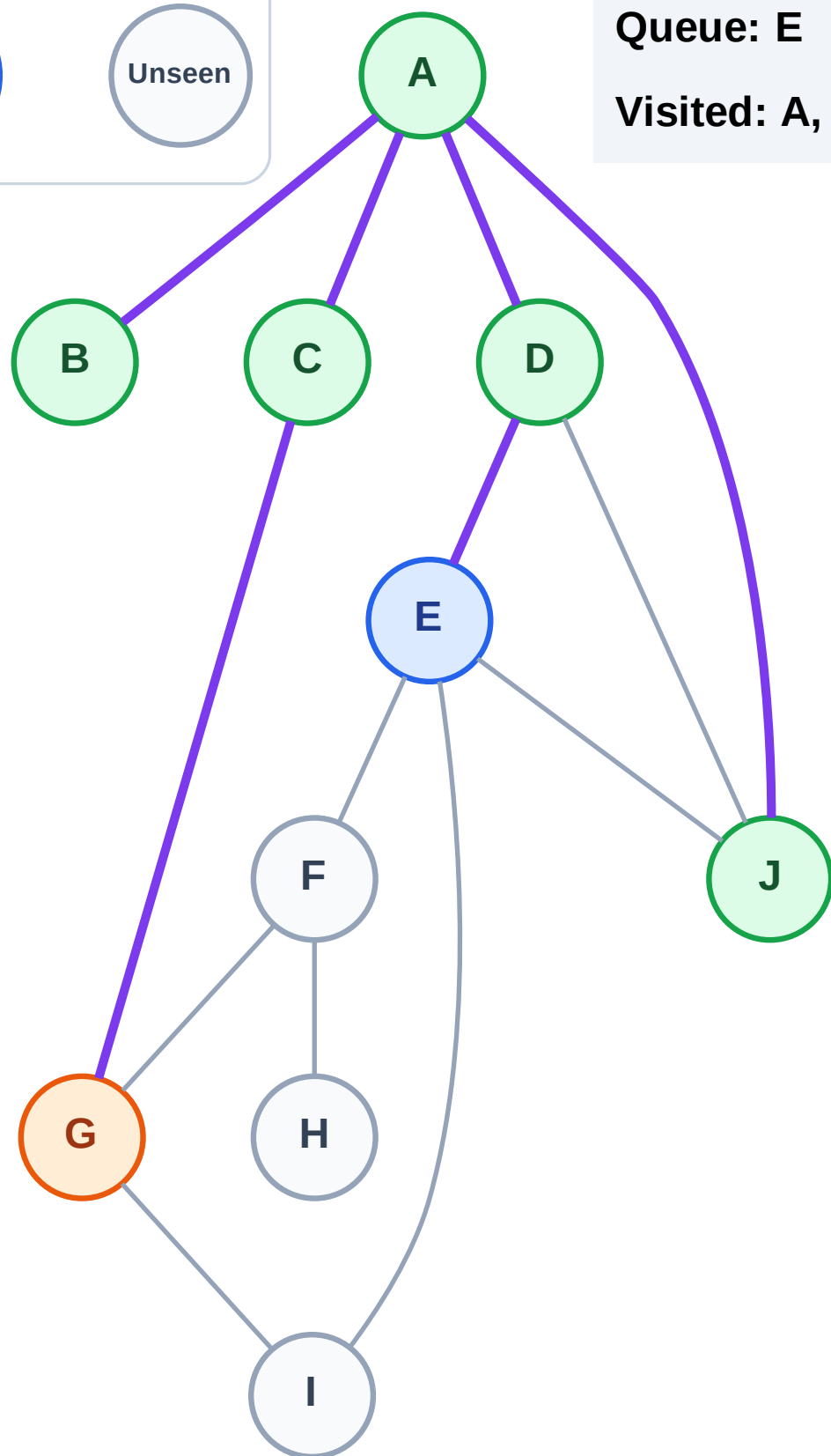
Step 12: Process node G

Legend



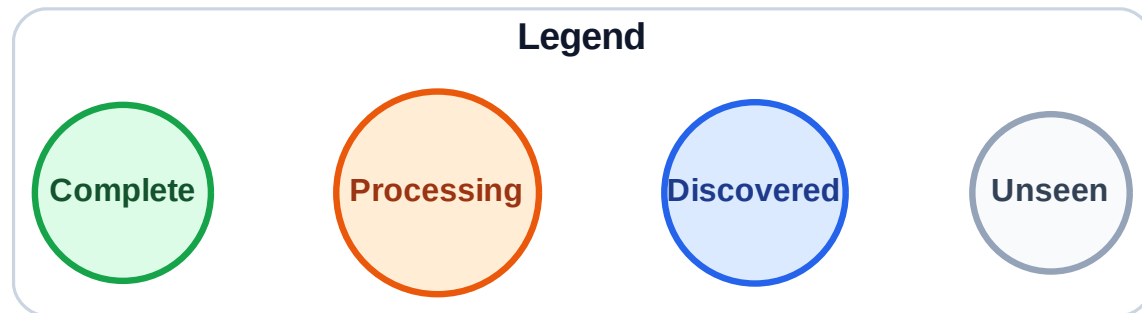
Queue: E

Visited: A, B, C, D, J

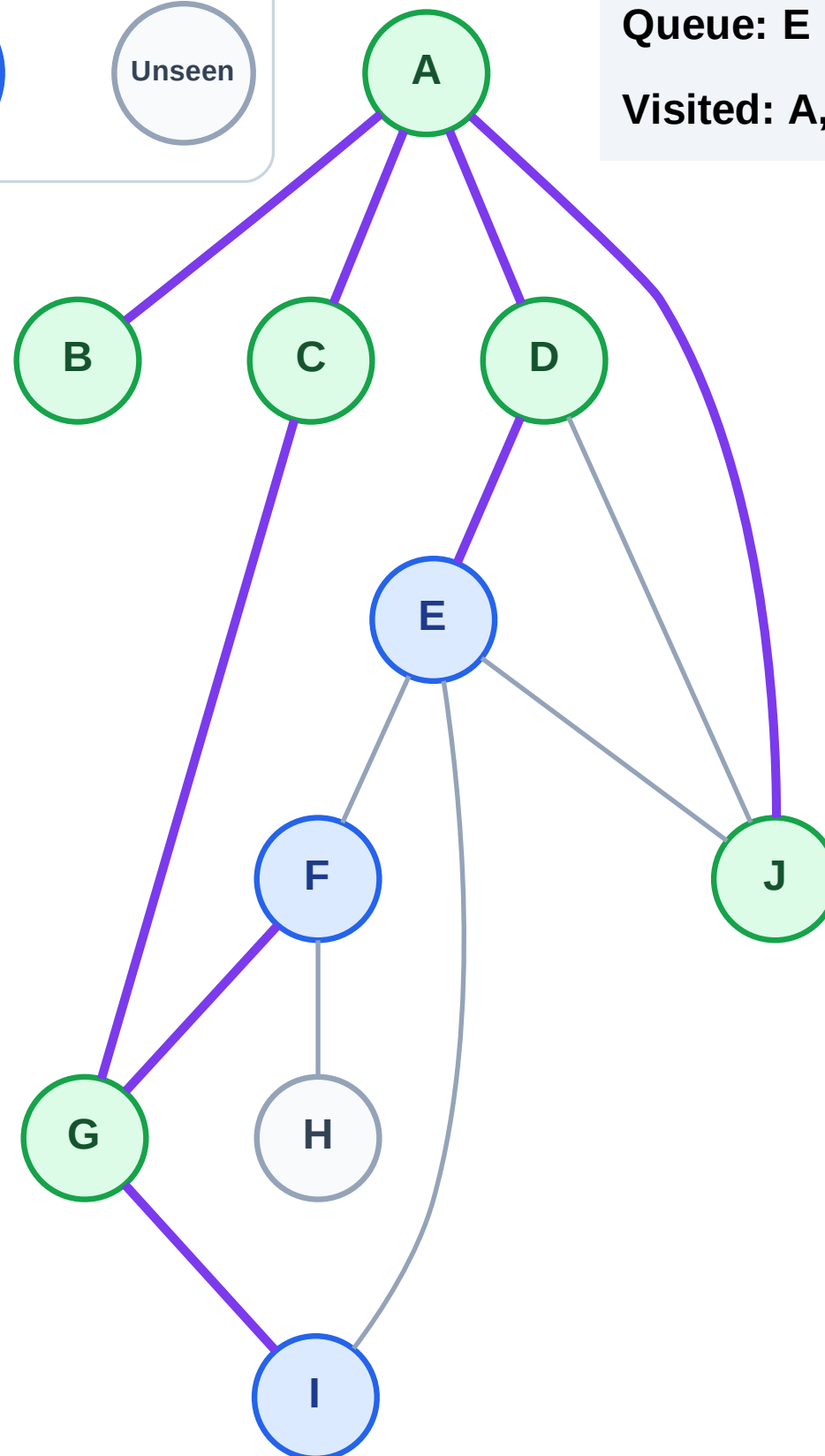


BFS Traversal

Step 13: Discover F, I from G



Queue: E → F → I
Visited: A, B, C, D, G, J



BFS Traversal

Step 14: Process node E

Legend

Complete

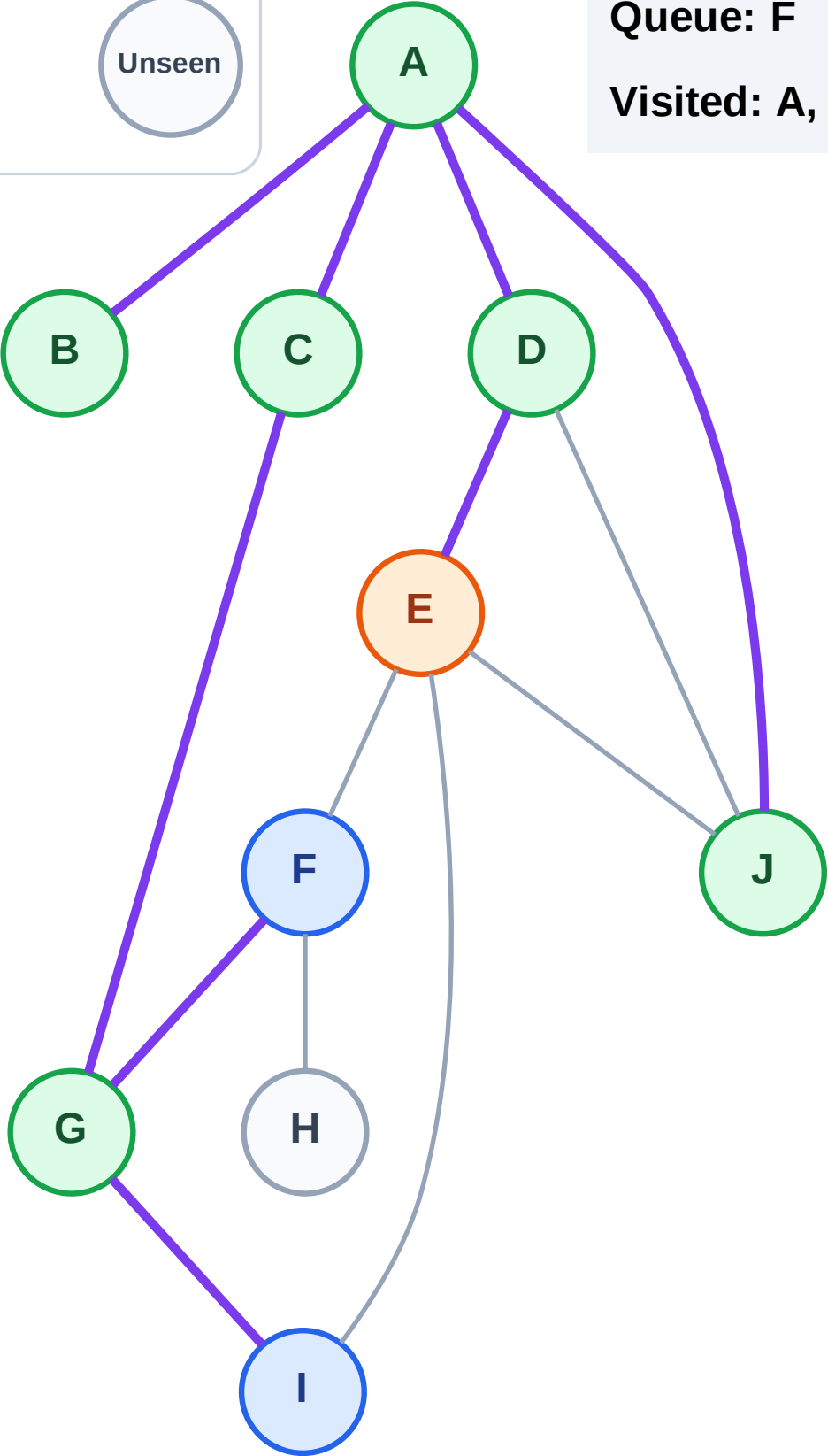
Processing

Discovered

Unseen

Queue: F → I

Visited: A, B, C, D, G, J



BFS Traversal

Step 15: No new neighbors from E

Legend

Complete

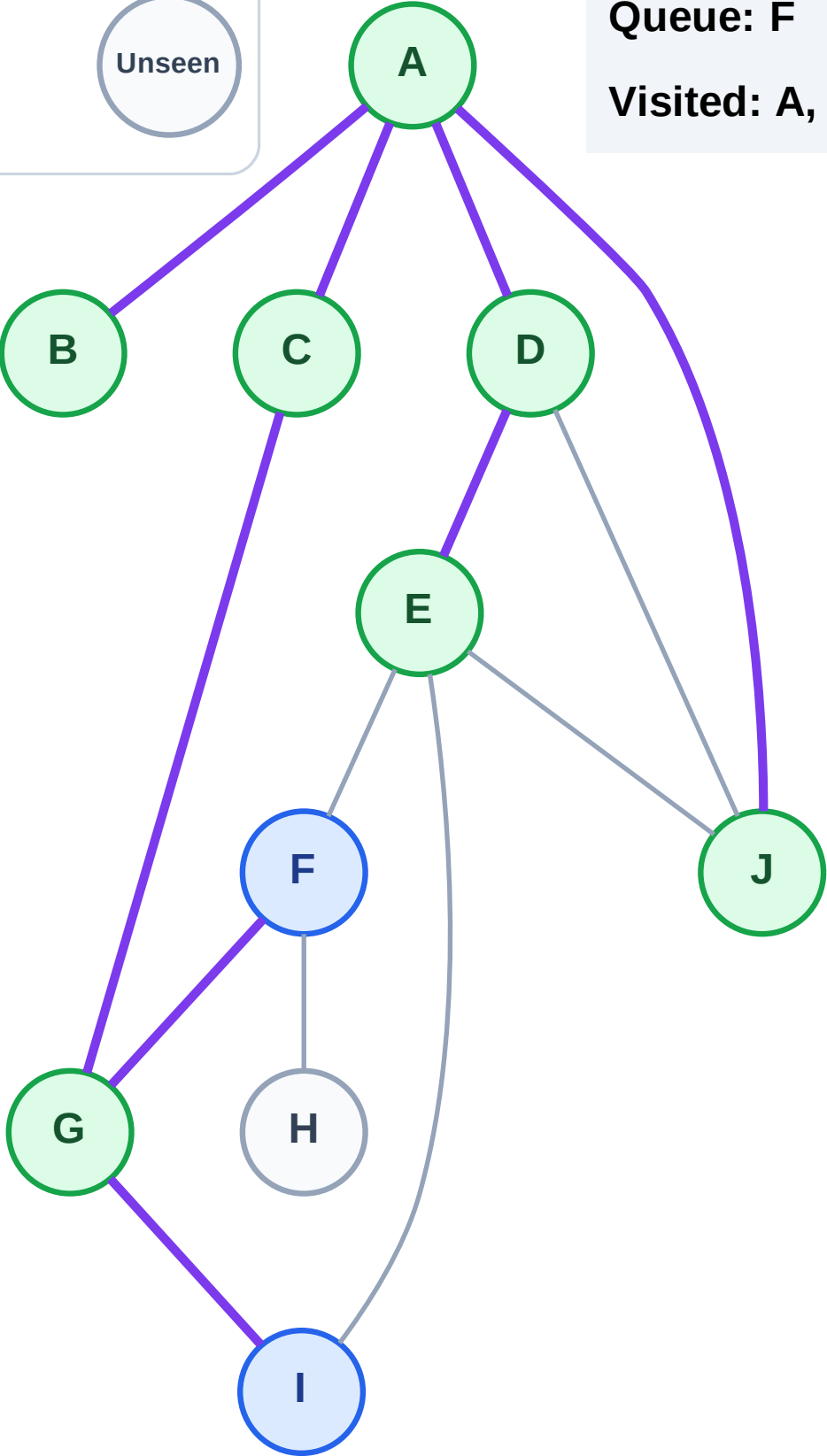
Processing

Discovered

Unseen

Queue: F → I

Visited: A, B, C, D, E, G, J



BFS Traversal

Step 16: Process node F

Legend

Complete

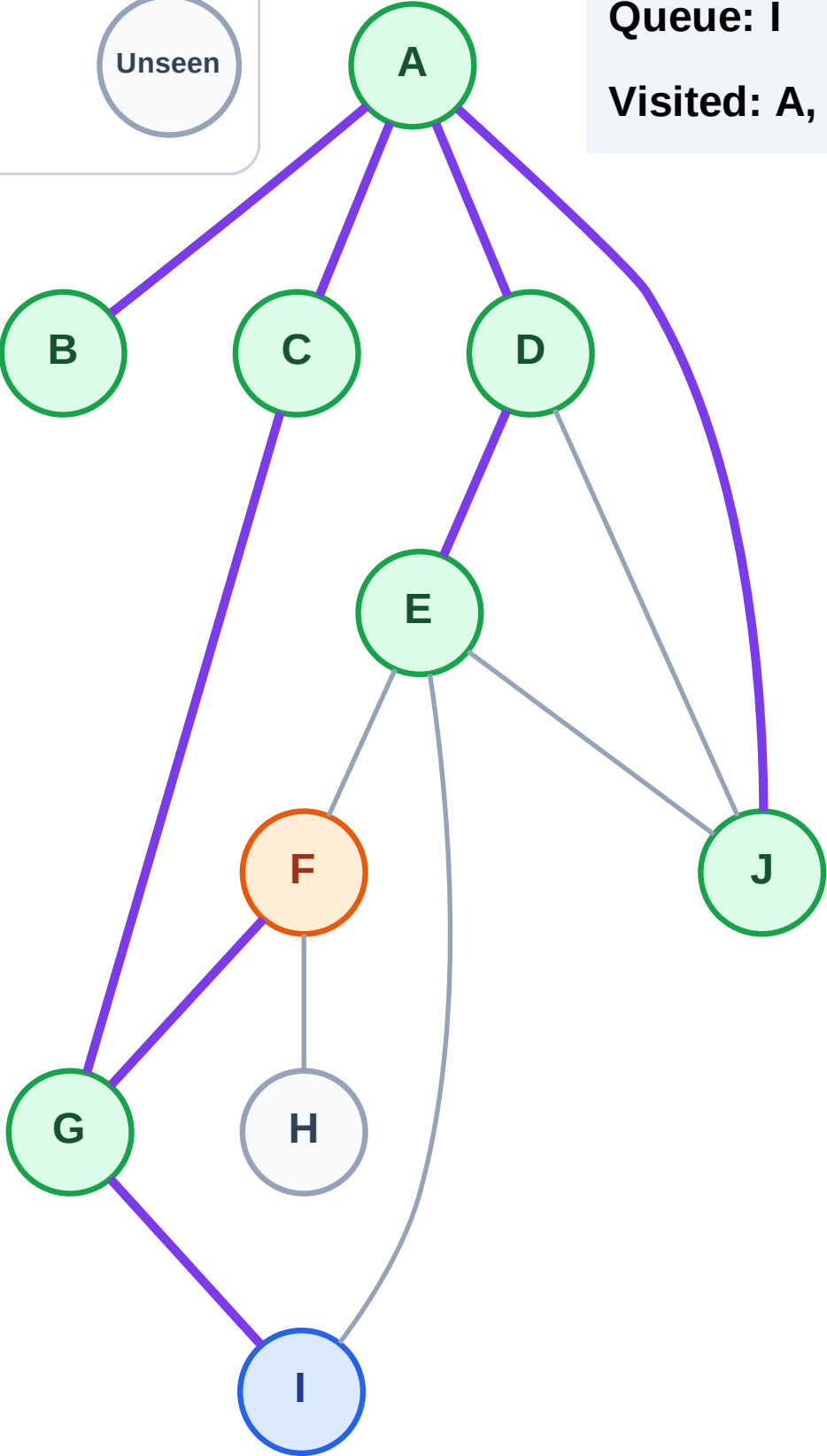
Processing

Discovered

Unseen

Queue: I

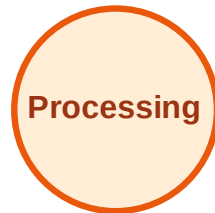
Visited: A, B, C, D, E, G, J



BFS Traversal

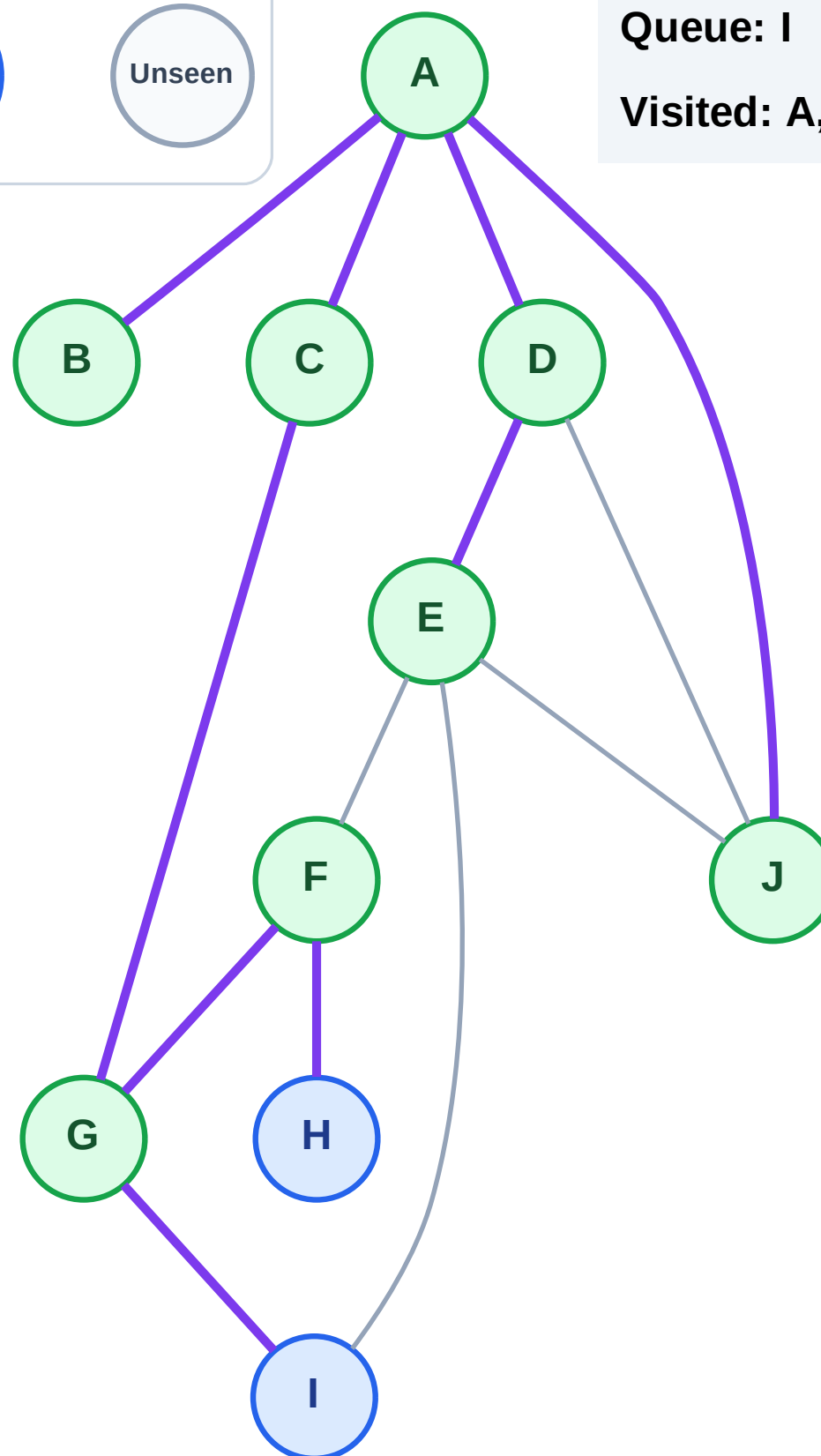
Step 17: Discover H from F

Legend



Queue: I → H

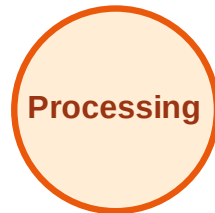
Visited: A, B, C, D, E, F, G, J



BFS Traversal

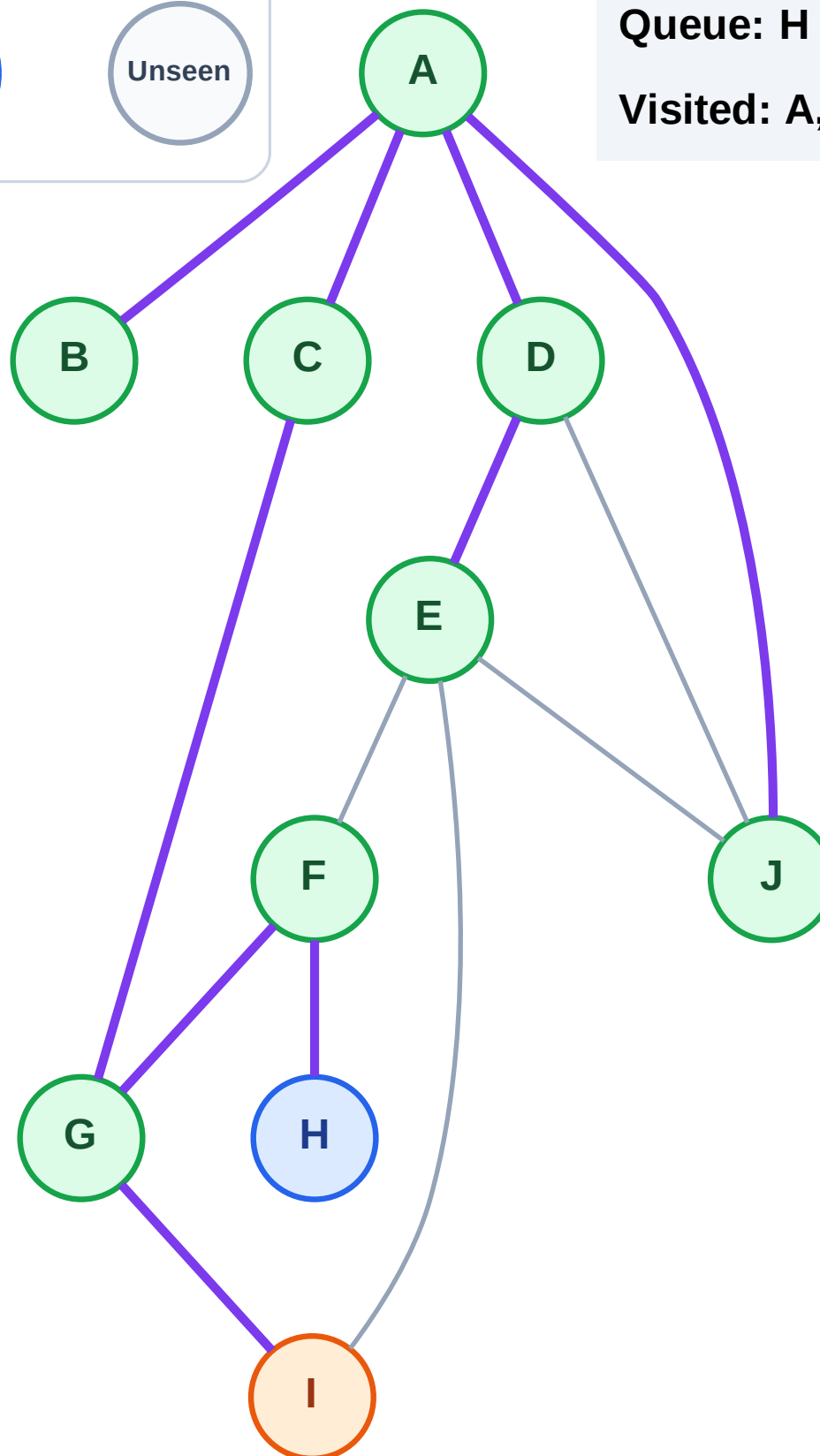
Step 18: Process node I

Legend



Queue: H

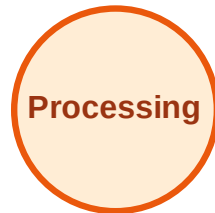
Visited: A, B, C, D, E, F, G, J



BFS Traversal

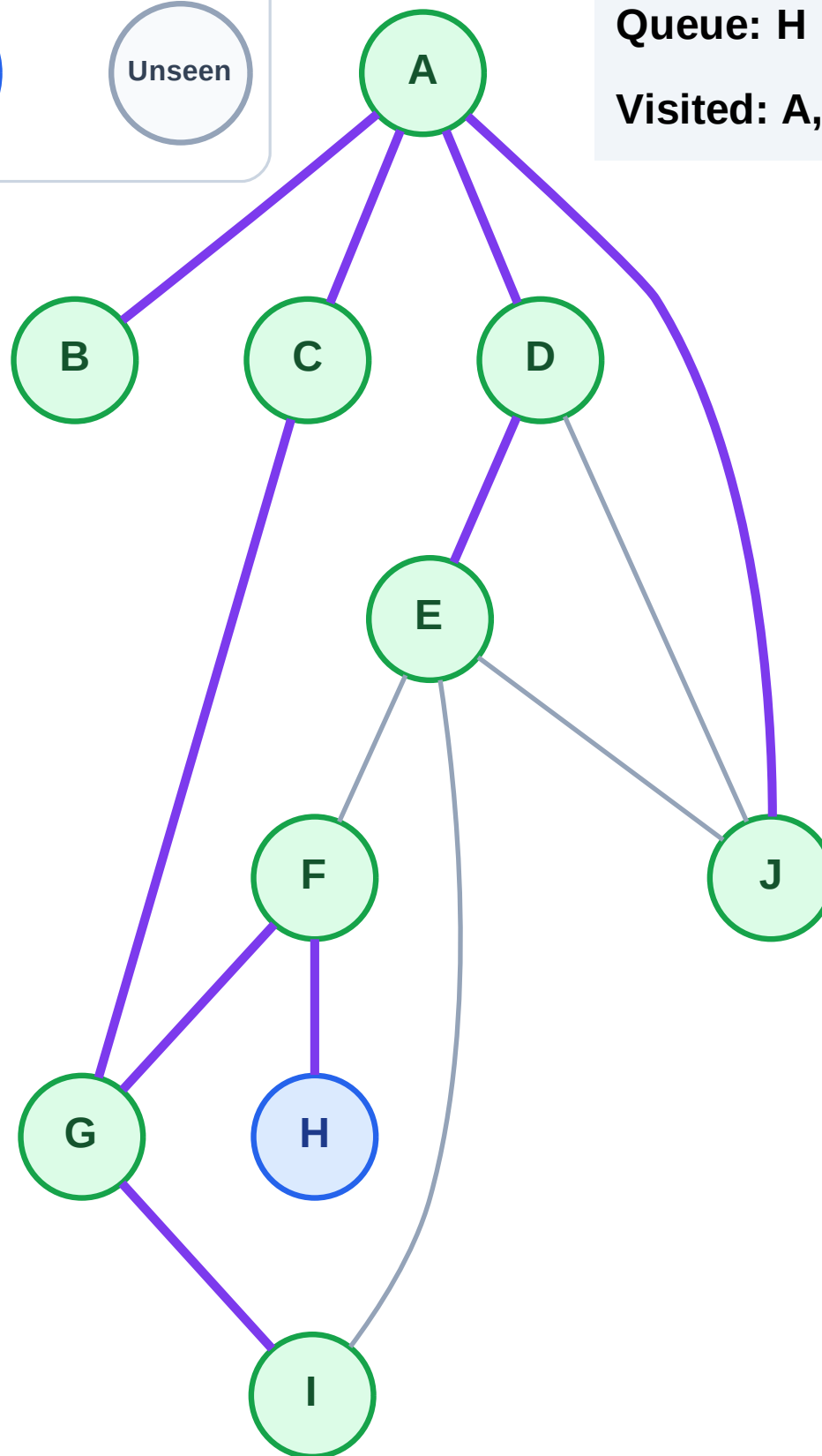
Step 19: No new neighbors from I

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 20: Process node H

Legend

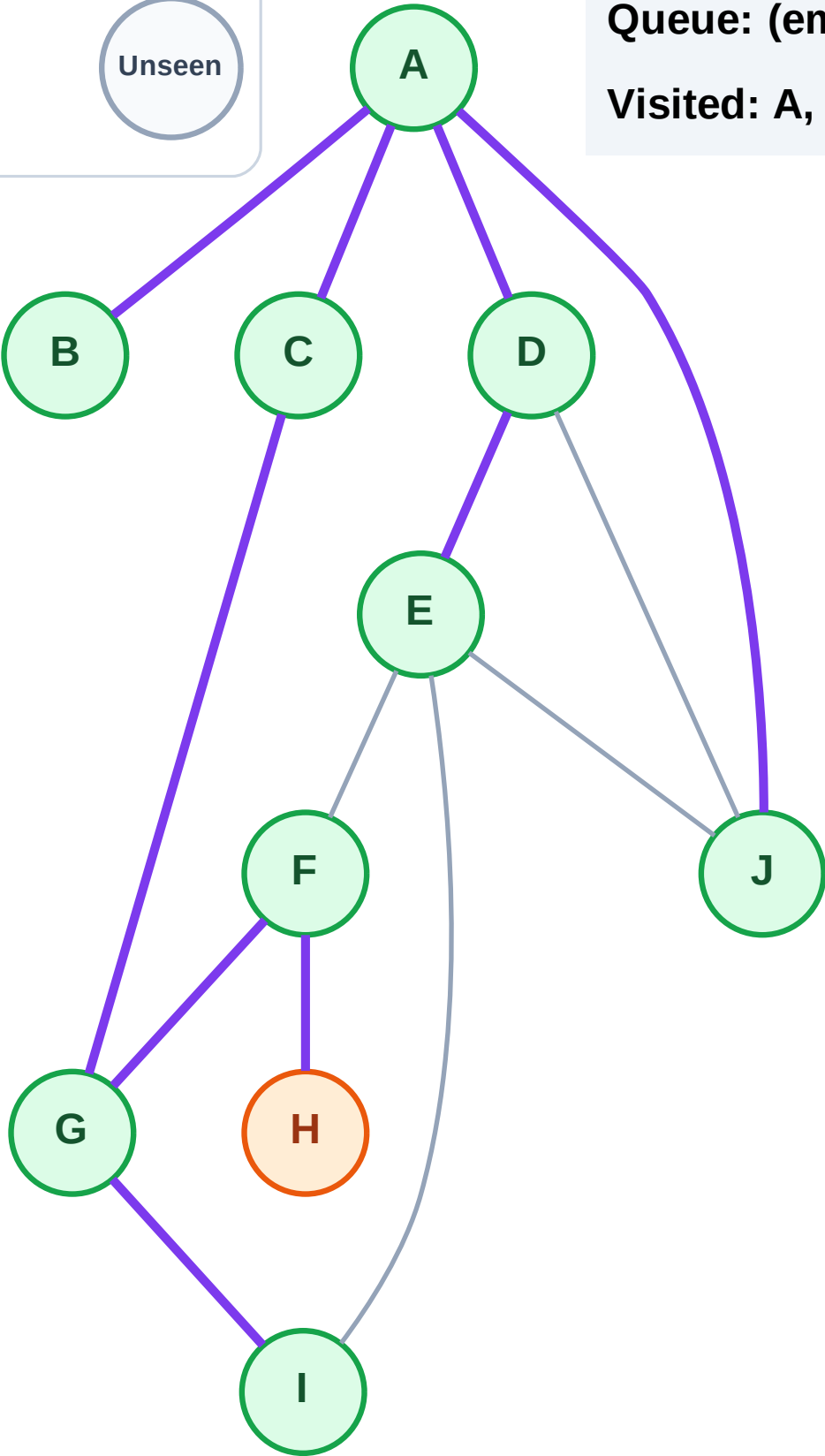
Complete

Processing

Discovered

Unseen

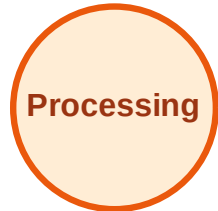
Queue: (empty)
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

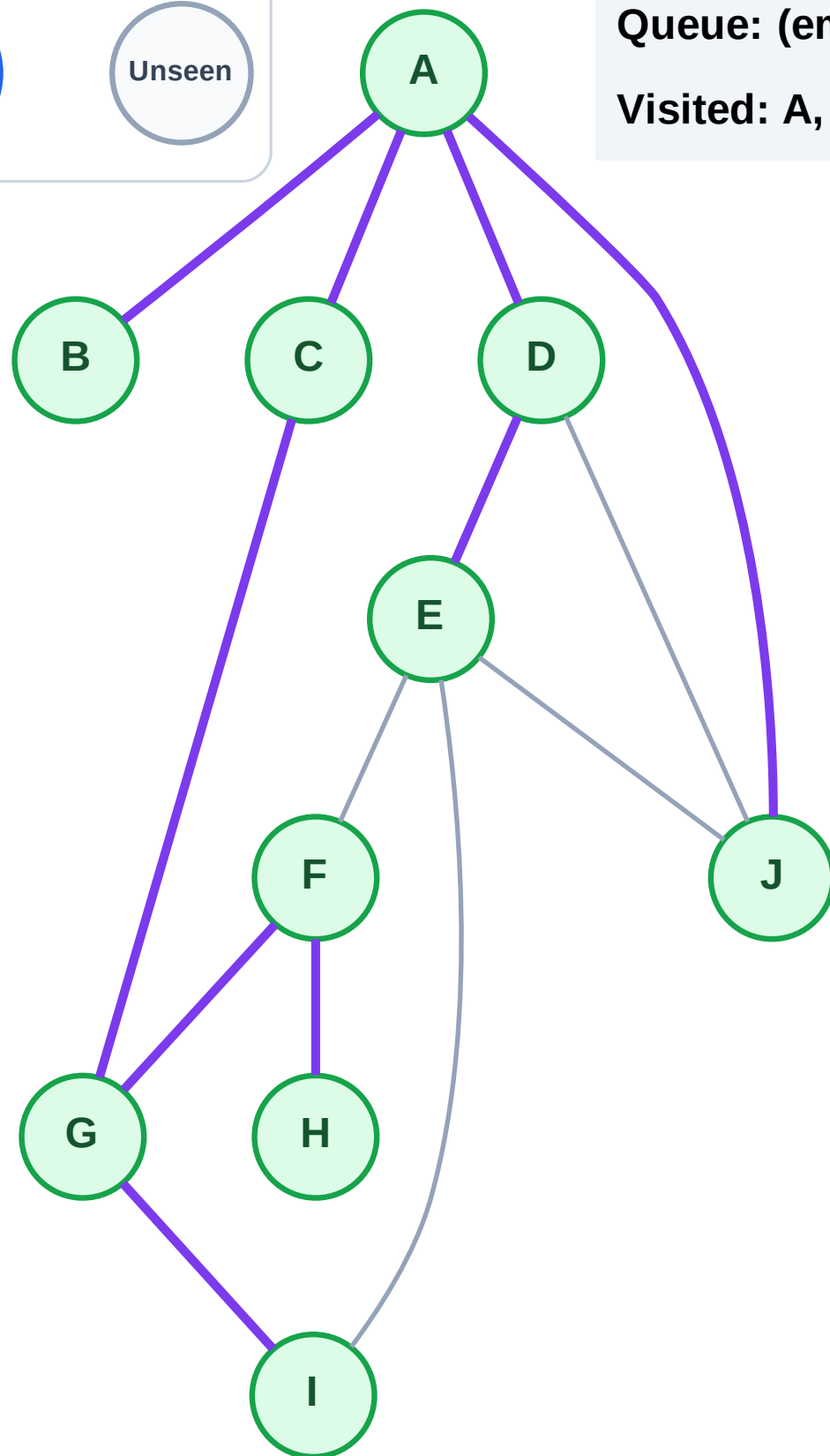
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

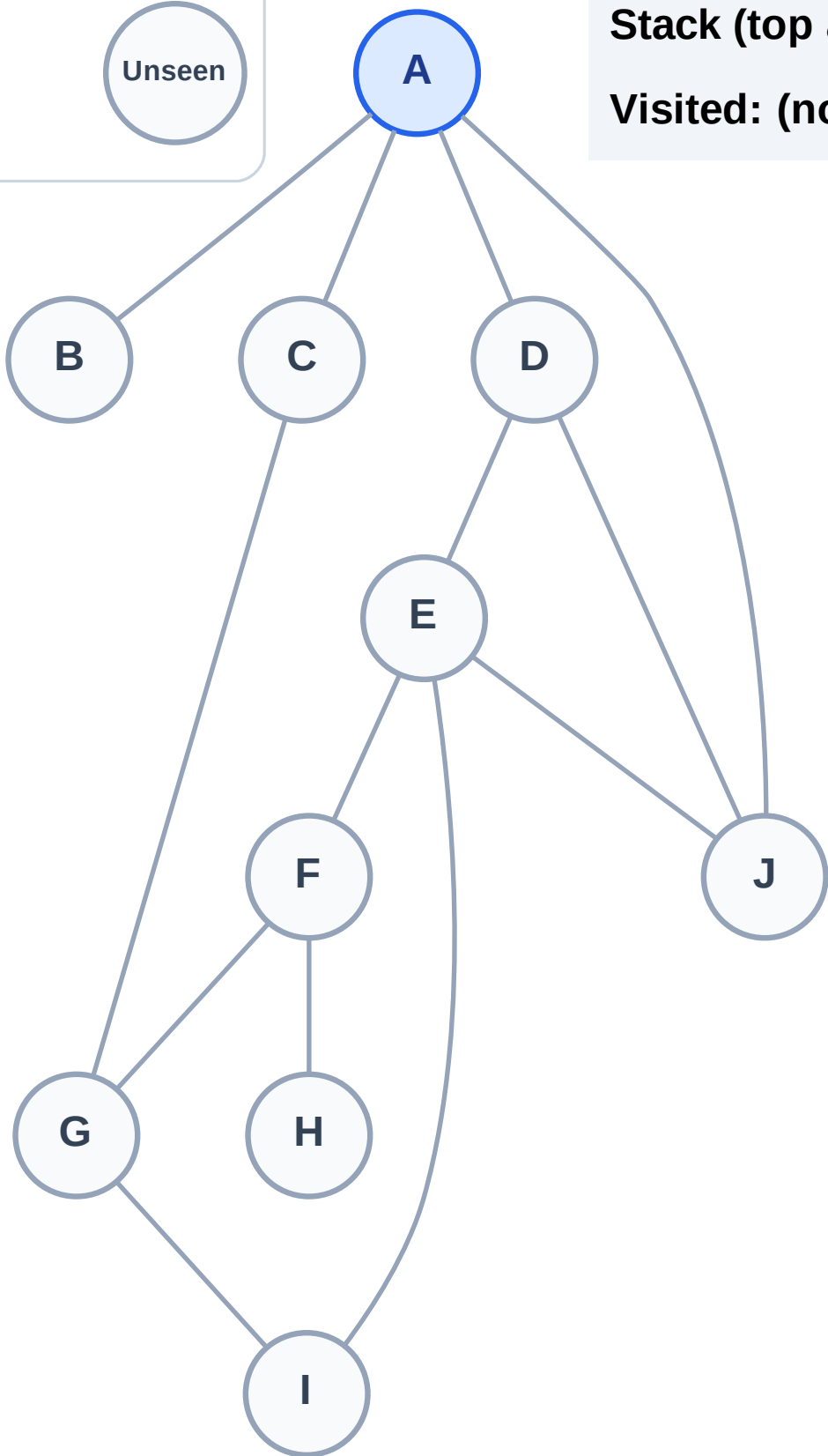
Step 1: Start at A

Legend



Stack (top at right): A

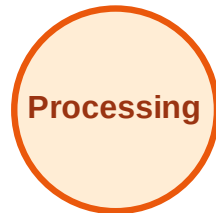
Visited: (none)



DFS Traversal

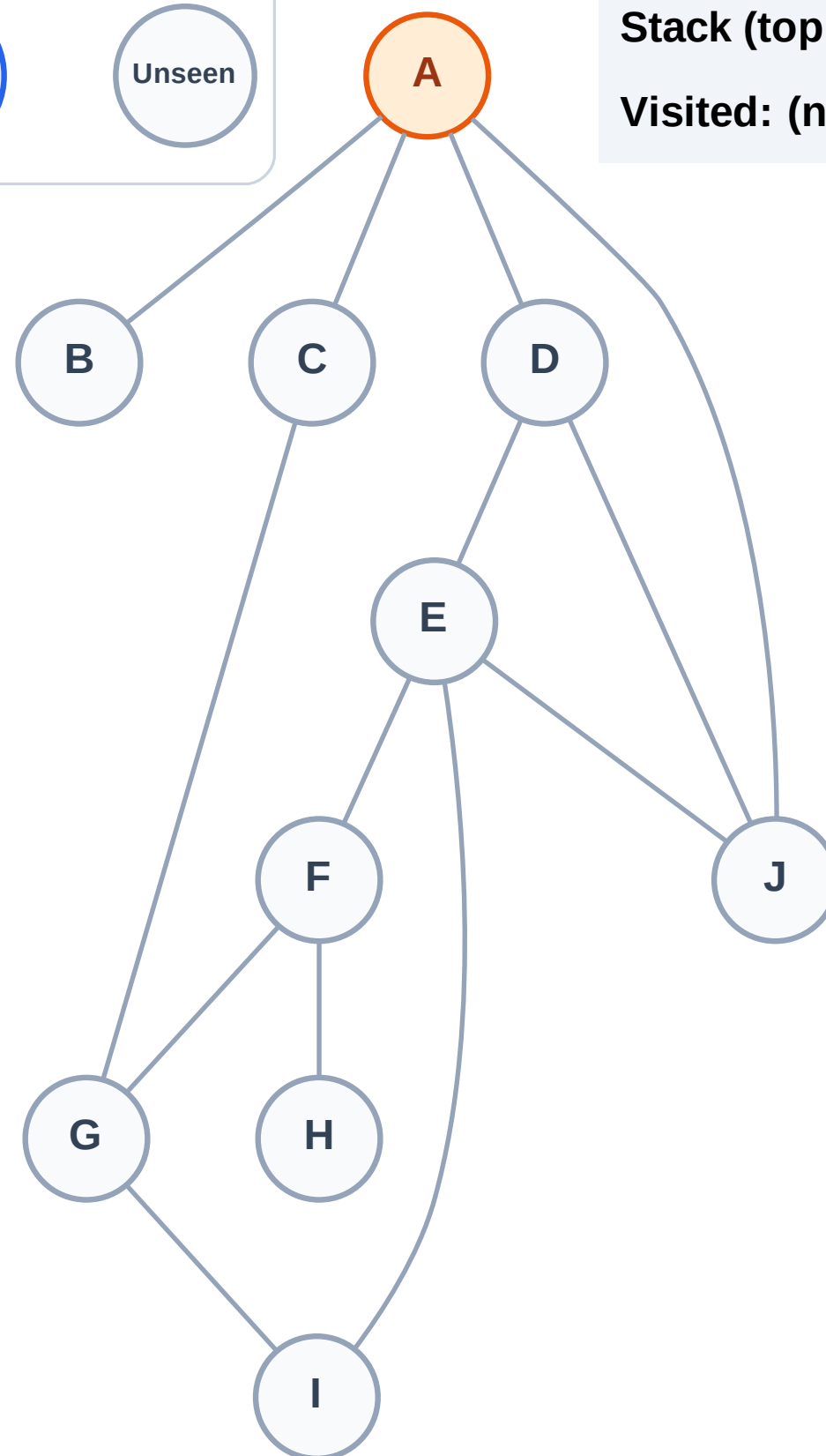
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, D, C, B from A

Legend

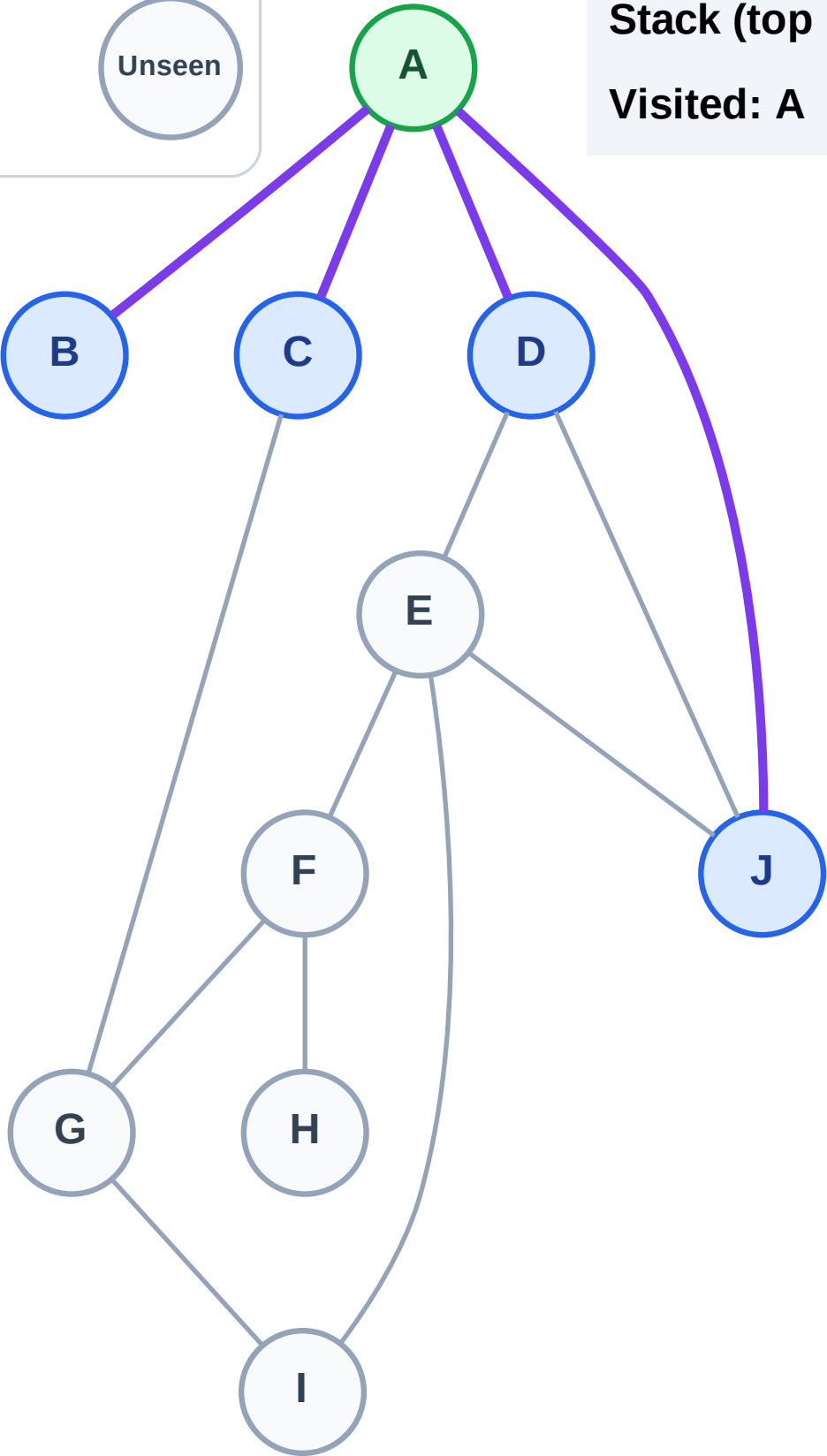
Complete

Processing

Discovered

Unseen

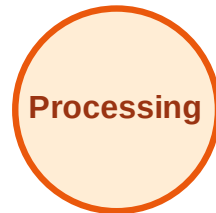
Stack (top at right): J | D | C | B
Visited: A



DFS Traversal

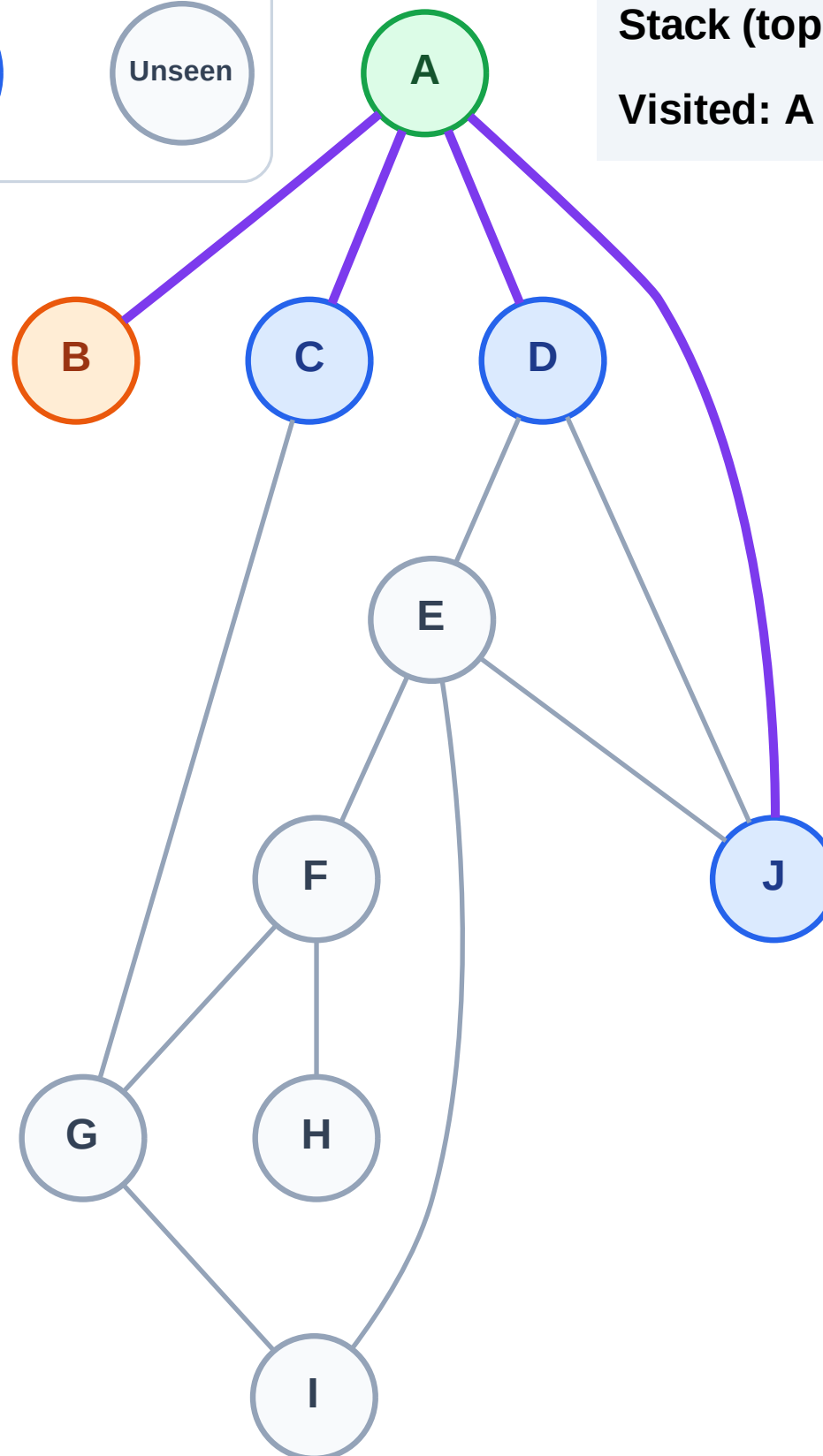
Step 4: Process node B

Legend



Stack (top at right): J | D | C

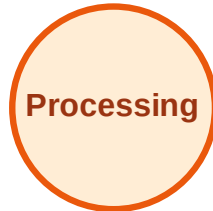
Visited: A



DFS Traversal

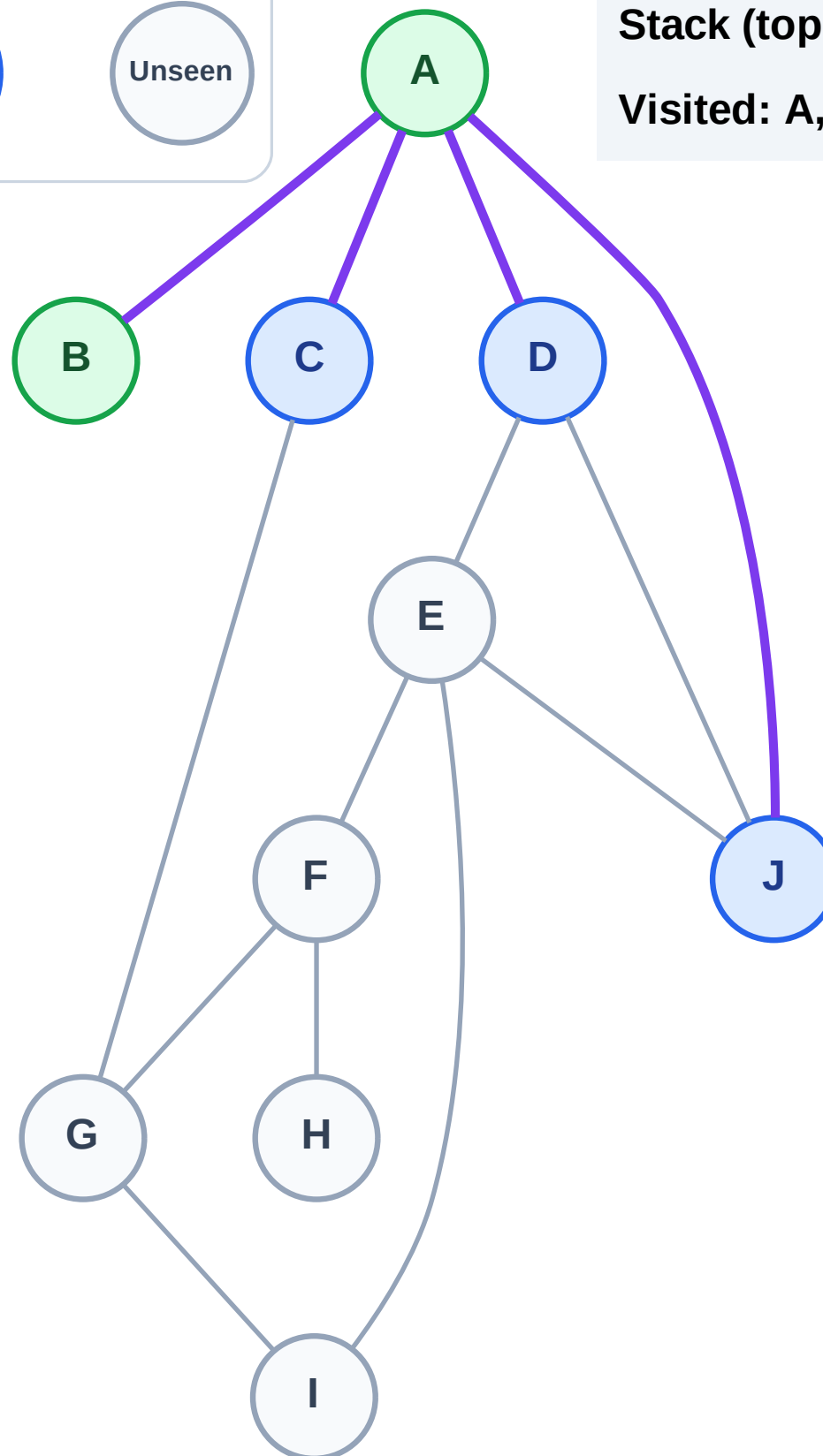
Step 5: No new neighbors from B

Legend



Stack (top at right): J | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

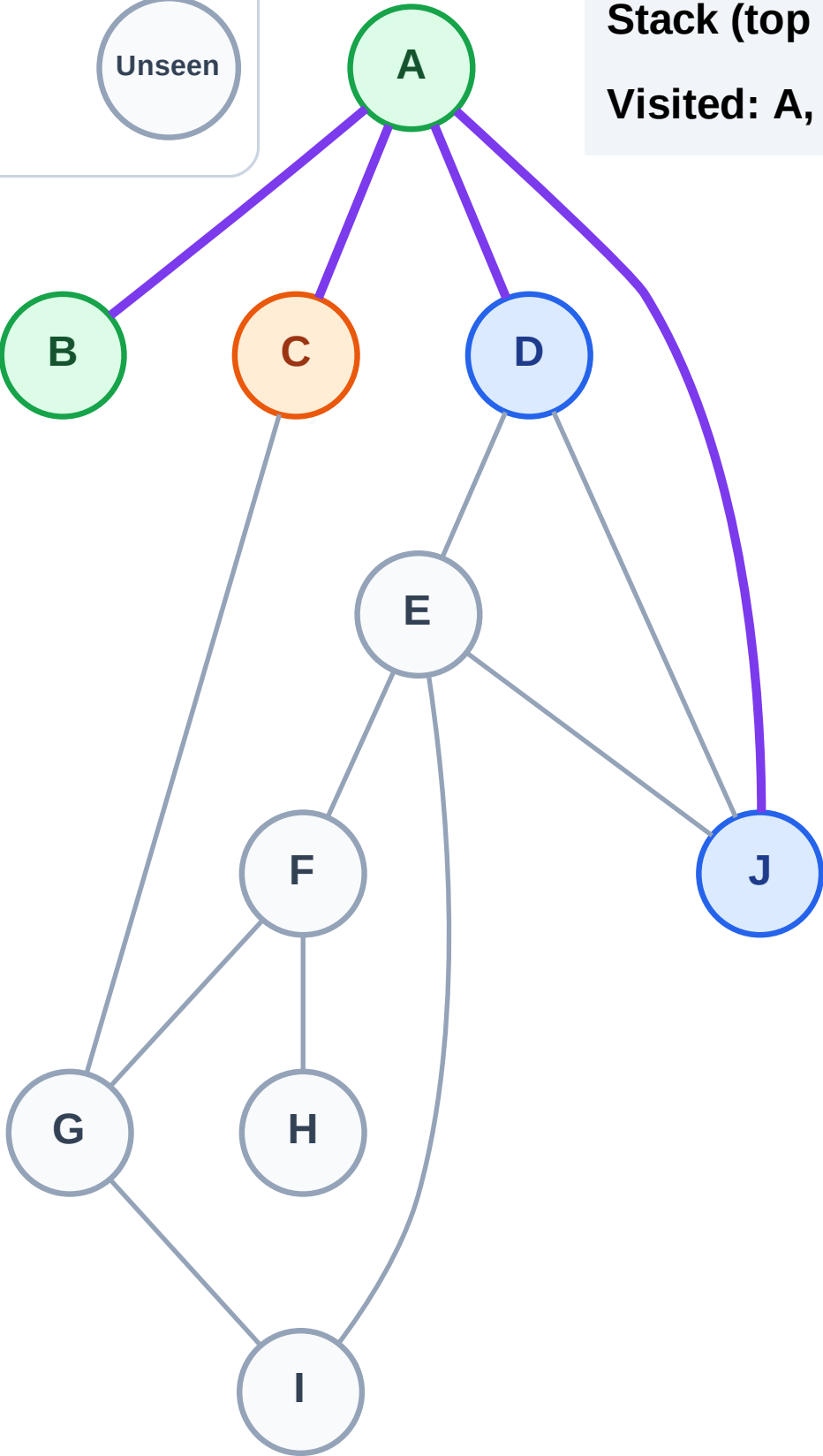
Complete

Processing

Discovered

Unseen

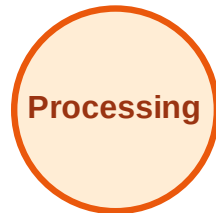
Stack (top at right): J | D
Visited: A, B



DFS Traversal

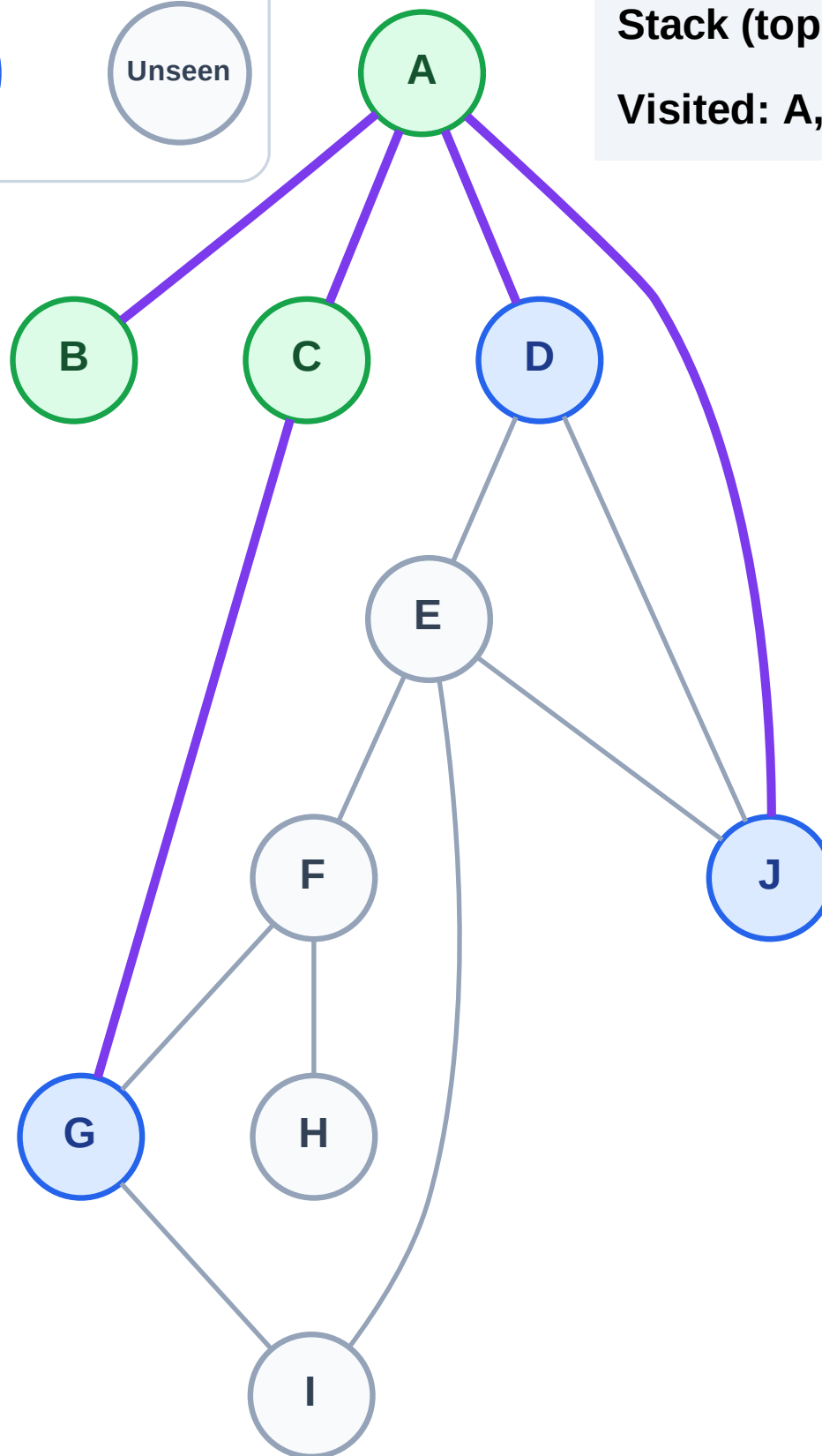
Step 7: Discover G from C

Legend



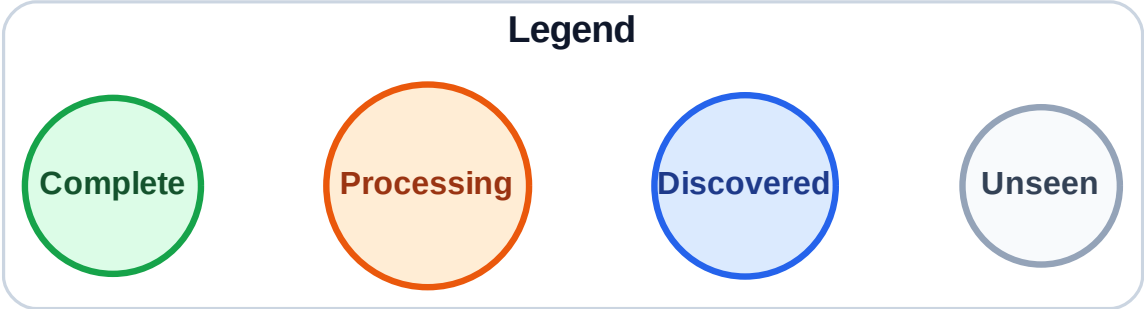
Stack (top at right): J | D | G

Visited: A, B, C

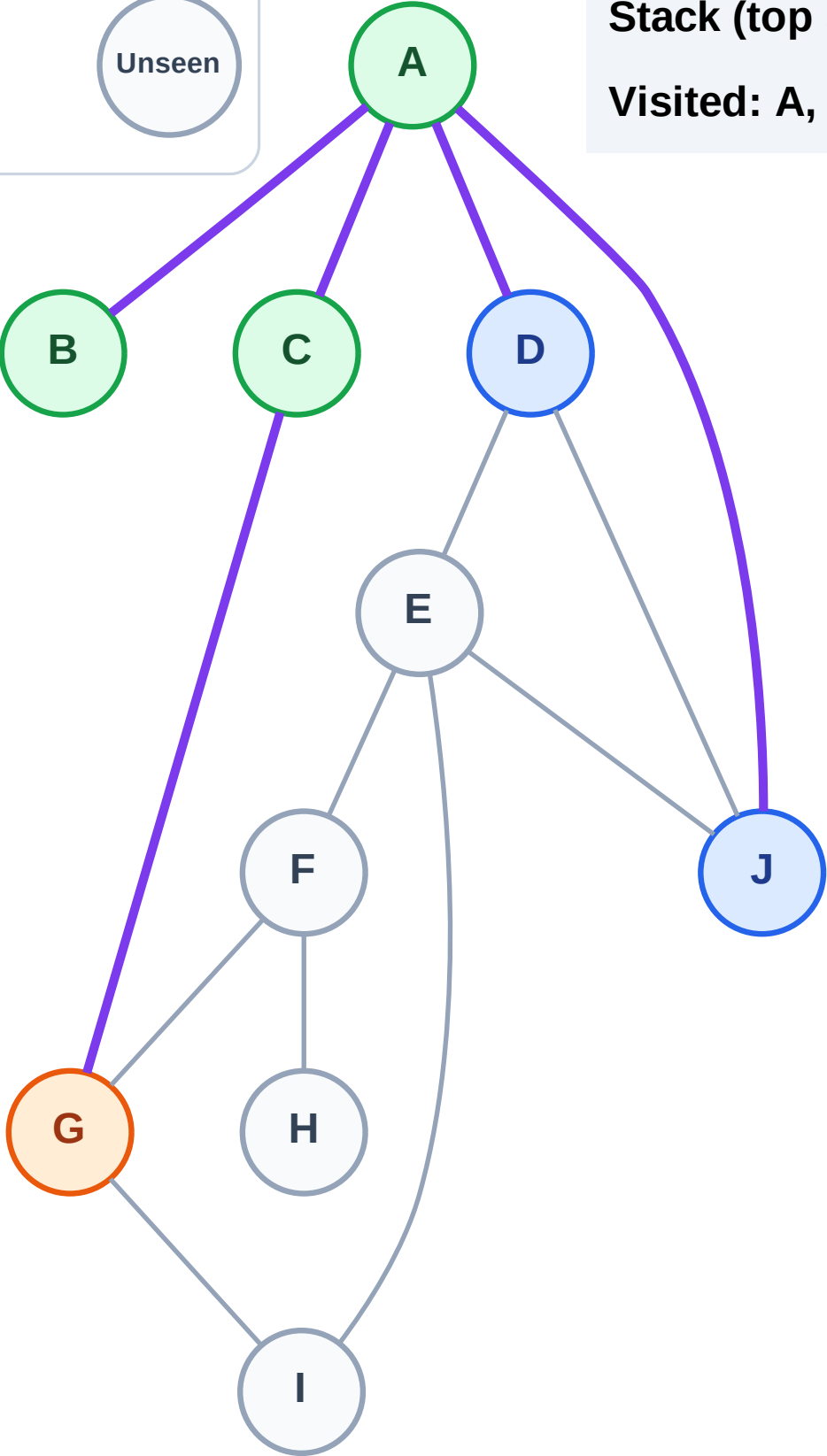


DFS Traversal

Step 8: Process node G



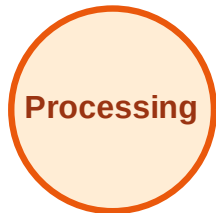
Stack (top at right): J | D
Visited: A, B, C



DFS Traversal

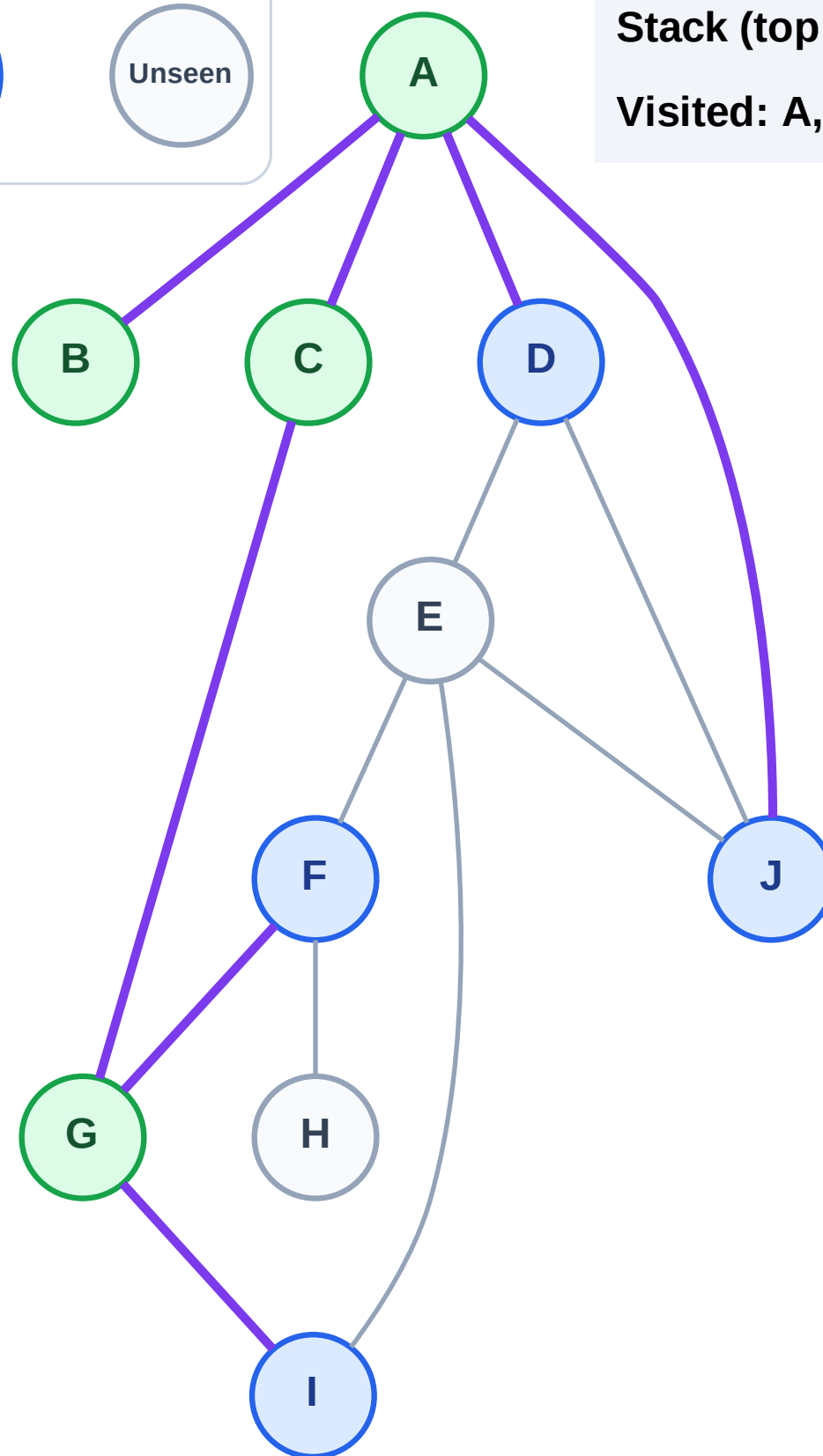
Step 9: Discover I, F from G

Legend



Stack (top at right): J | D | I | F

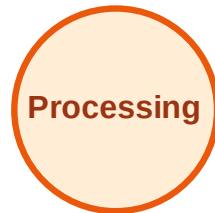
Visited: A, B, C, G



DFS Traversal

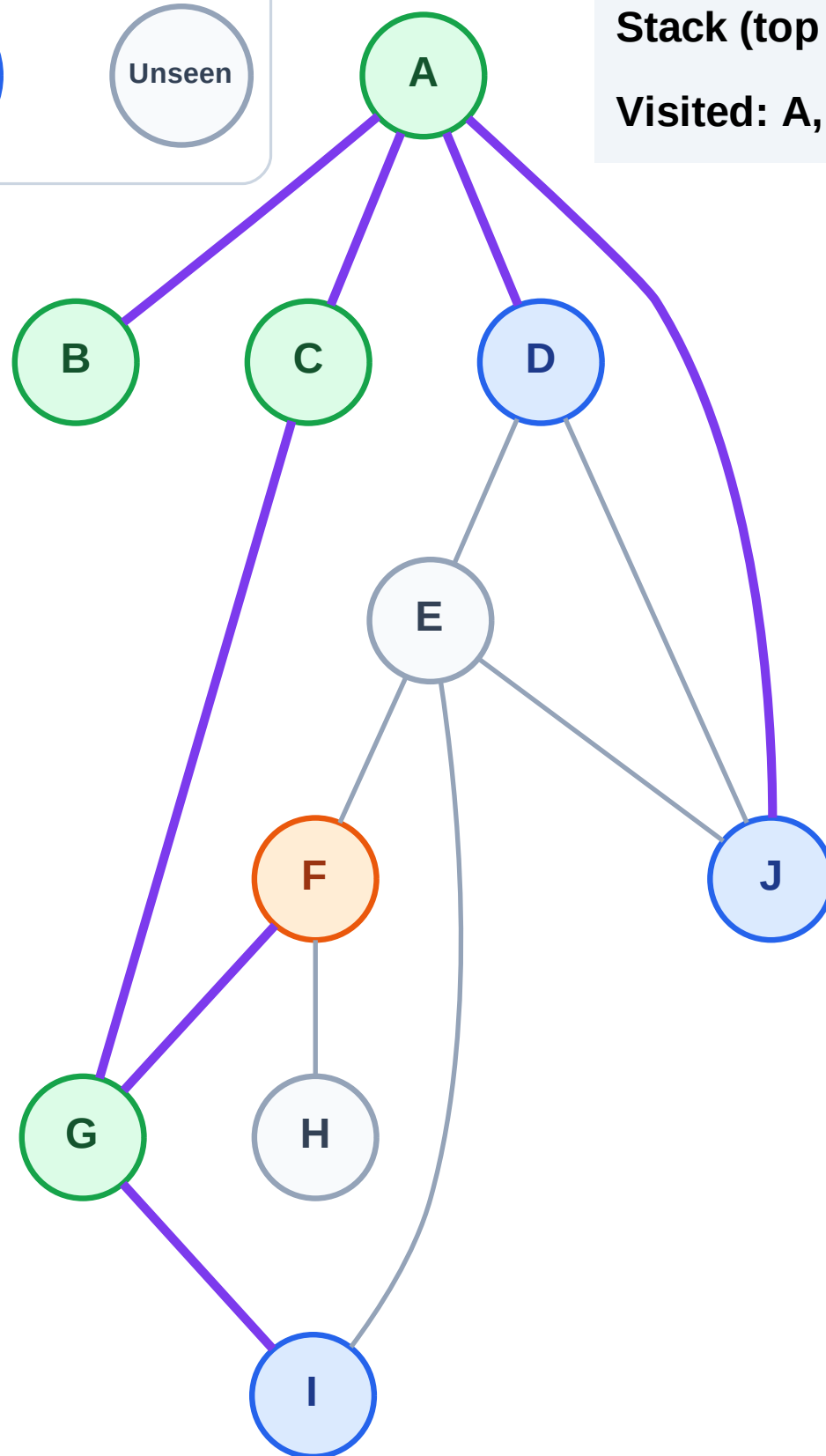
Step 10: Process node F

Legend



Stack (top at right): J | D | I

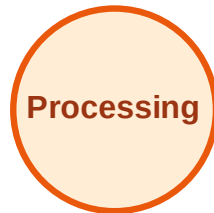
Visited: A, B, C, G



DFS Traversal

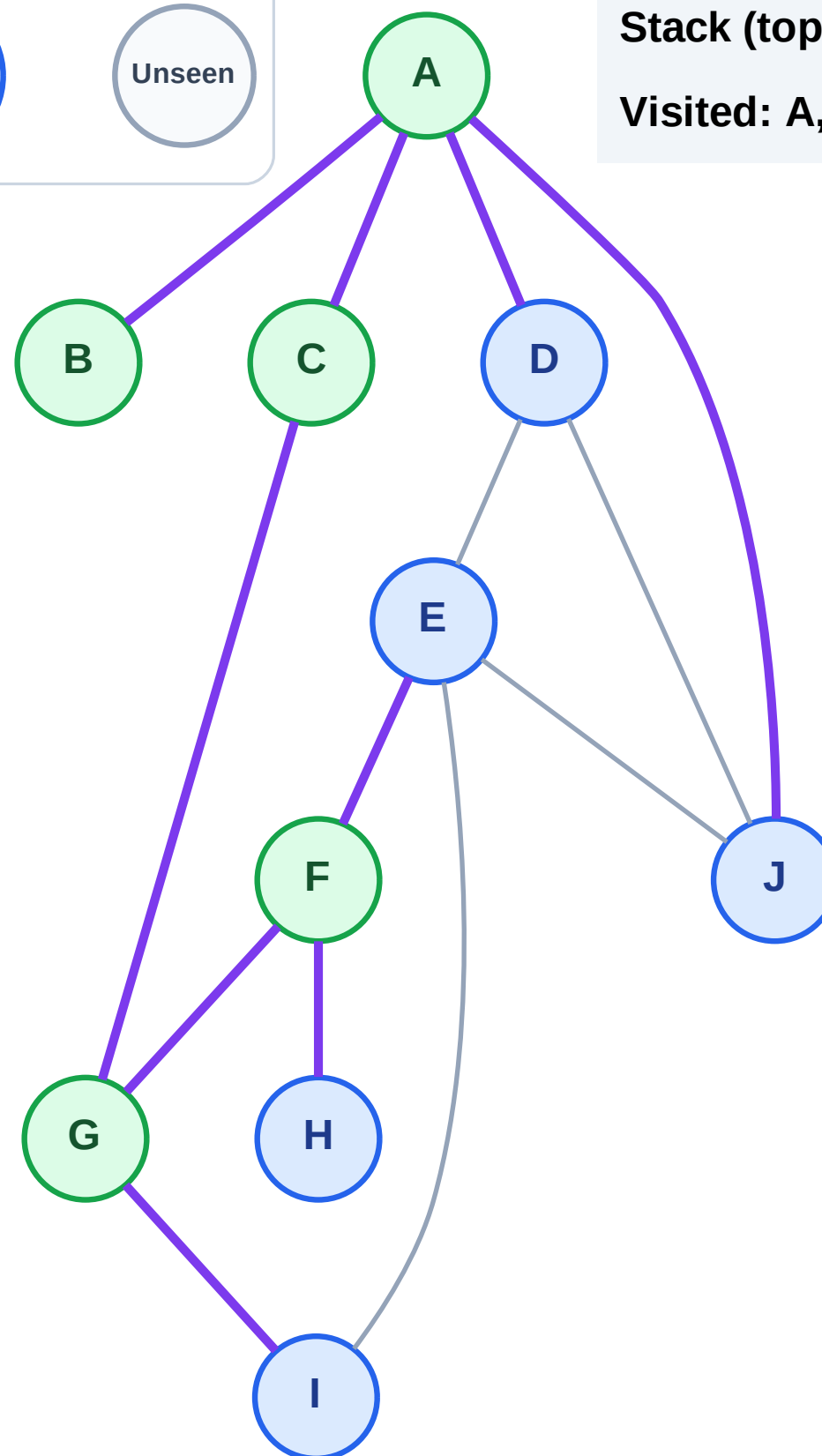
Step 11: Discover H, E from F

Legend



Stack (top at right): J | D | I | H | E

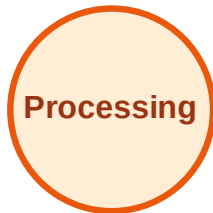
Visited: A, B, C, F, G



DFS Traversal

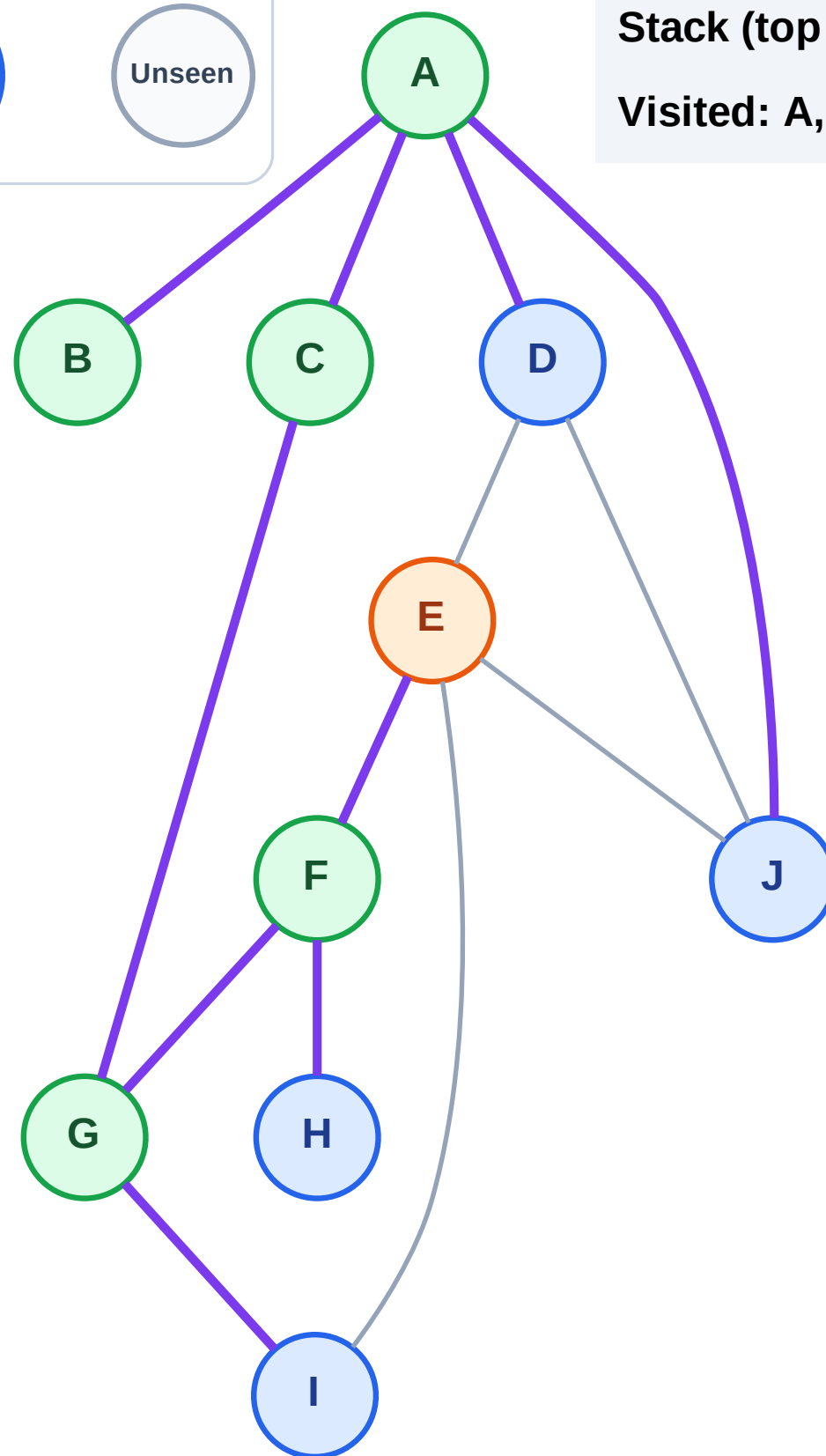
Step 12: Process node E

Legend



Stack (top at right): J | D | I | H

Visited: A, B, C, F, G



Step 13: No new neighbors from E

Legend



DFS Traversal

Step 14: Process node H

Legend

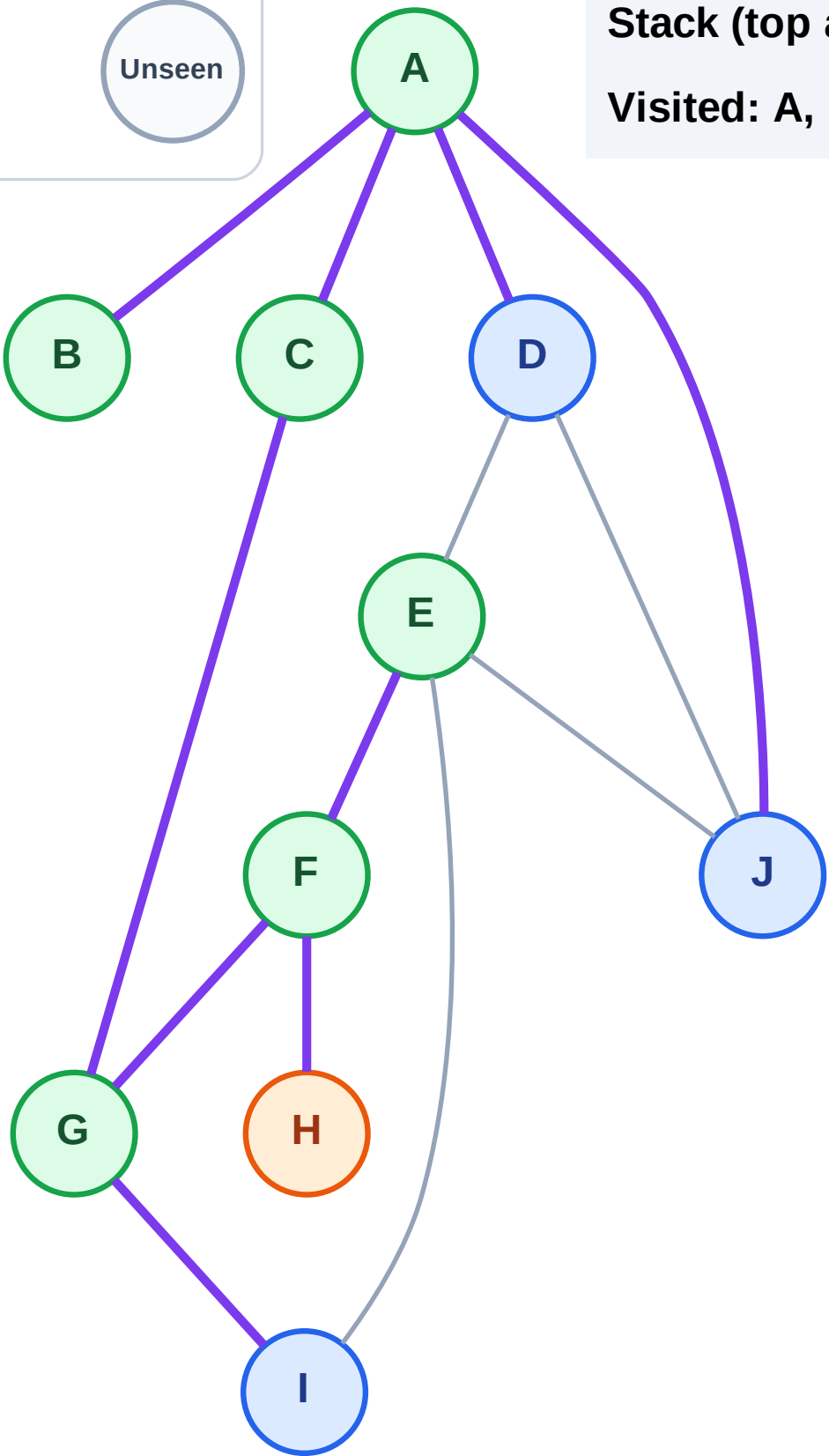
Complete

Processing

Discovered

Unseen

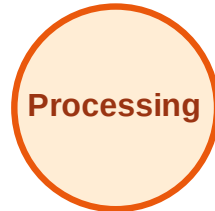
Stack (top at right): J | D | I
Visited: A, B, C, E, F, G



DFS Traversal

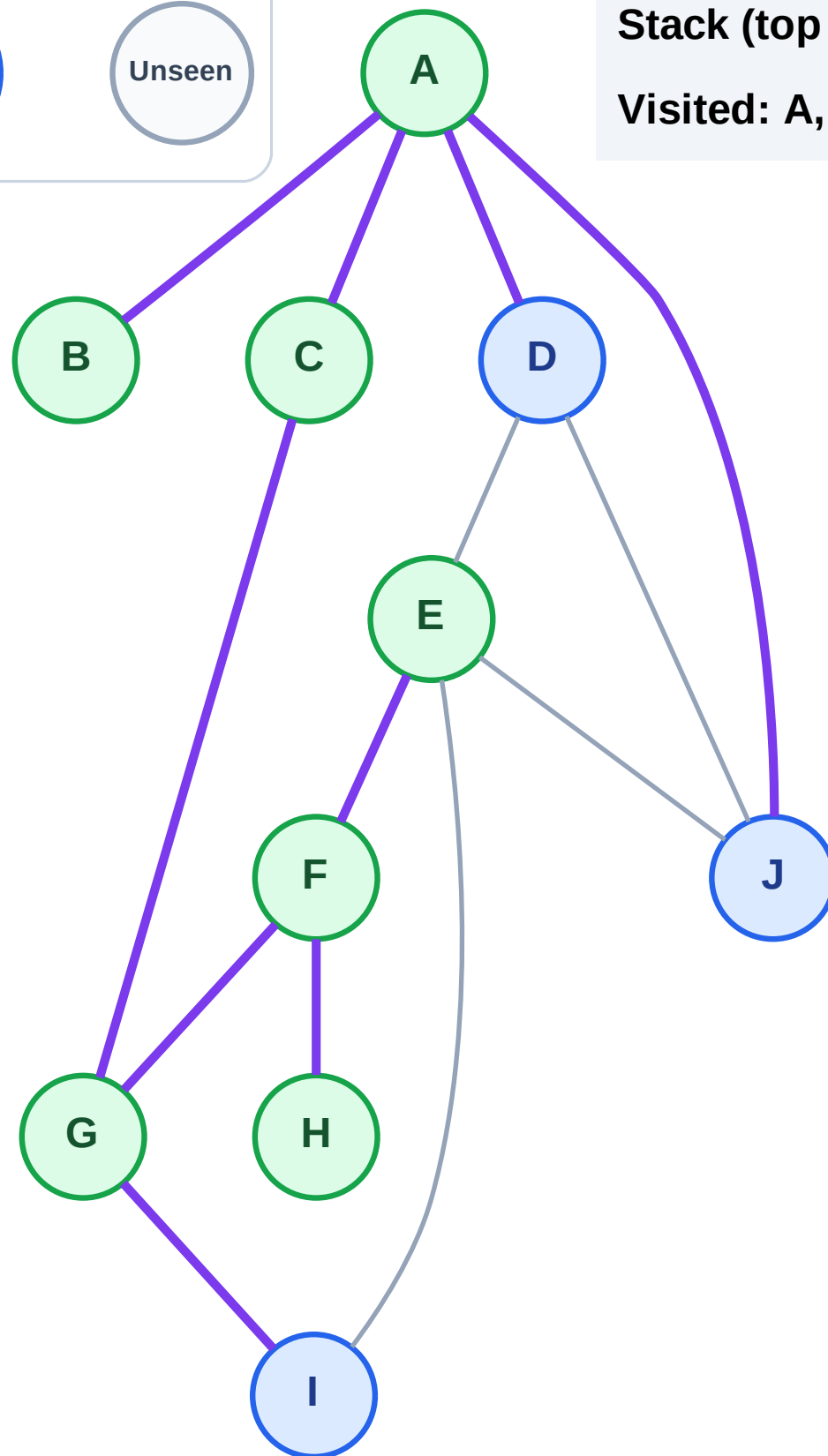
Step 15: No new neighbors from H

Legend



Stack (top at right): J | D | I

Visited: A, B, C, E, F, G, H



DFS Traversal

Step 16: Process node I

Legend

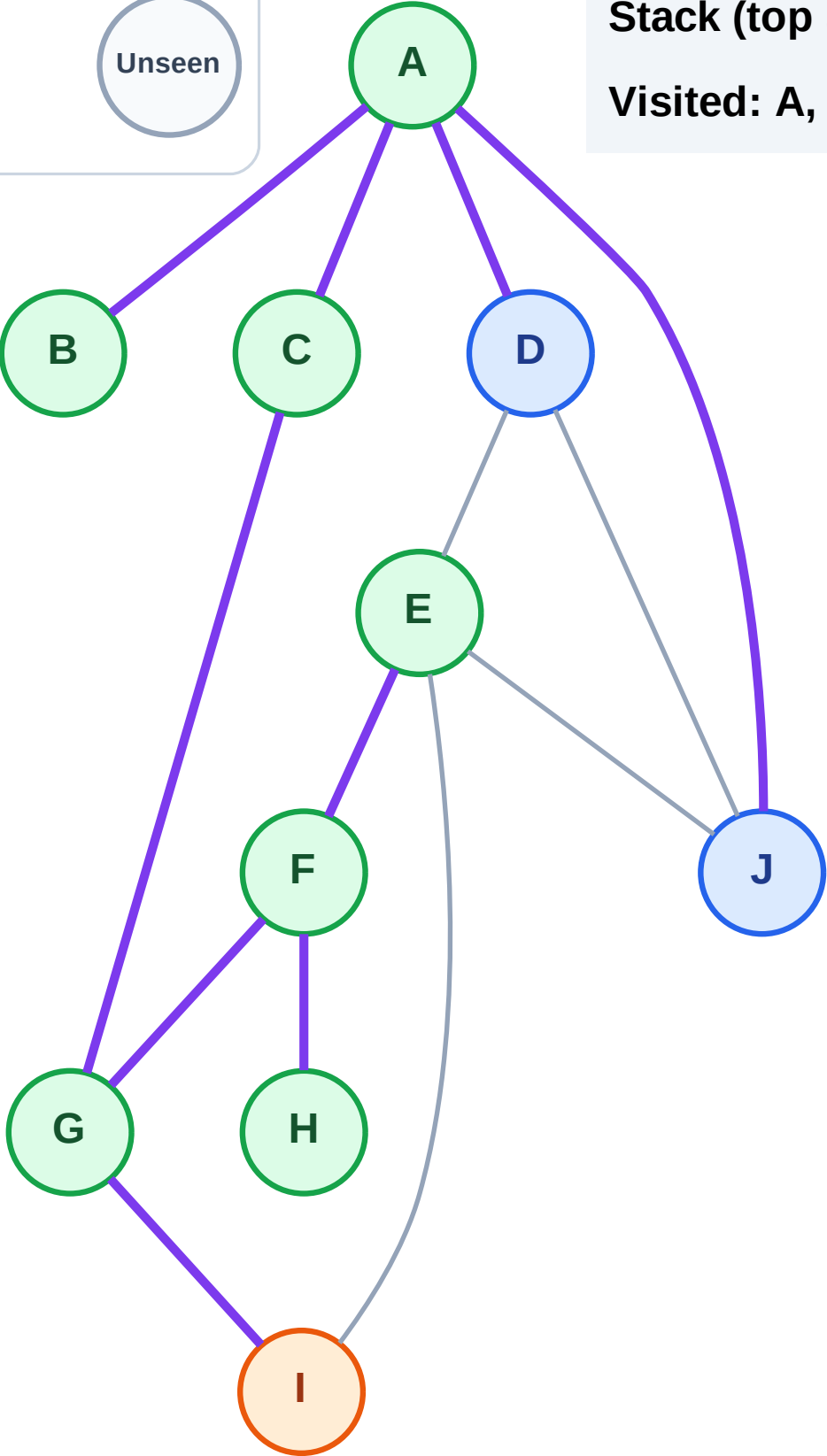
Complete

Processing

Discovered

Unseen

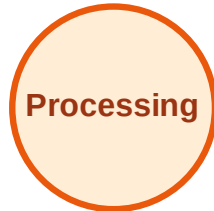
Stack (top at right): J | D
Visited: A, B, C, E, F, G, H



DFS Traversal

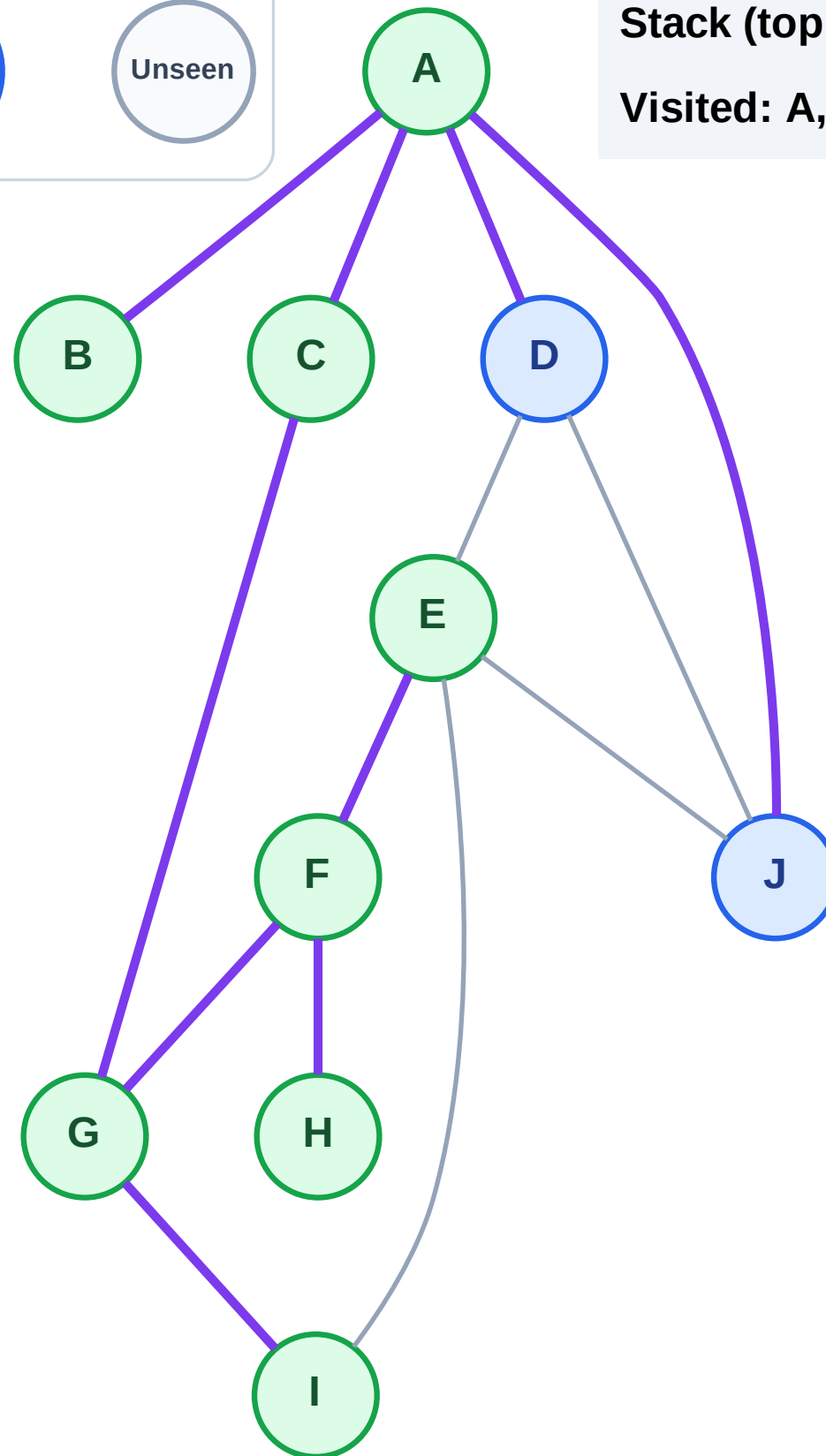
Step 17: No new neighbors from I

Legend



Stack (top at right): J | D

Visited: A, B, C, E, F, G, H, I



DFS Traversal

Step 18: Process node D

Legend

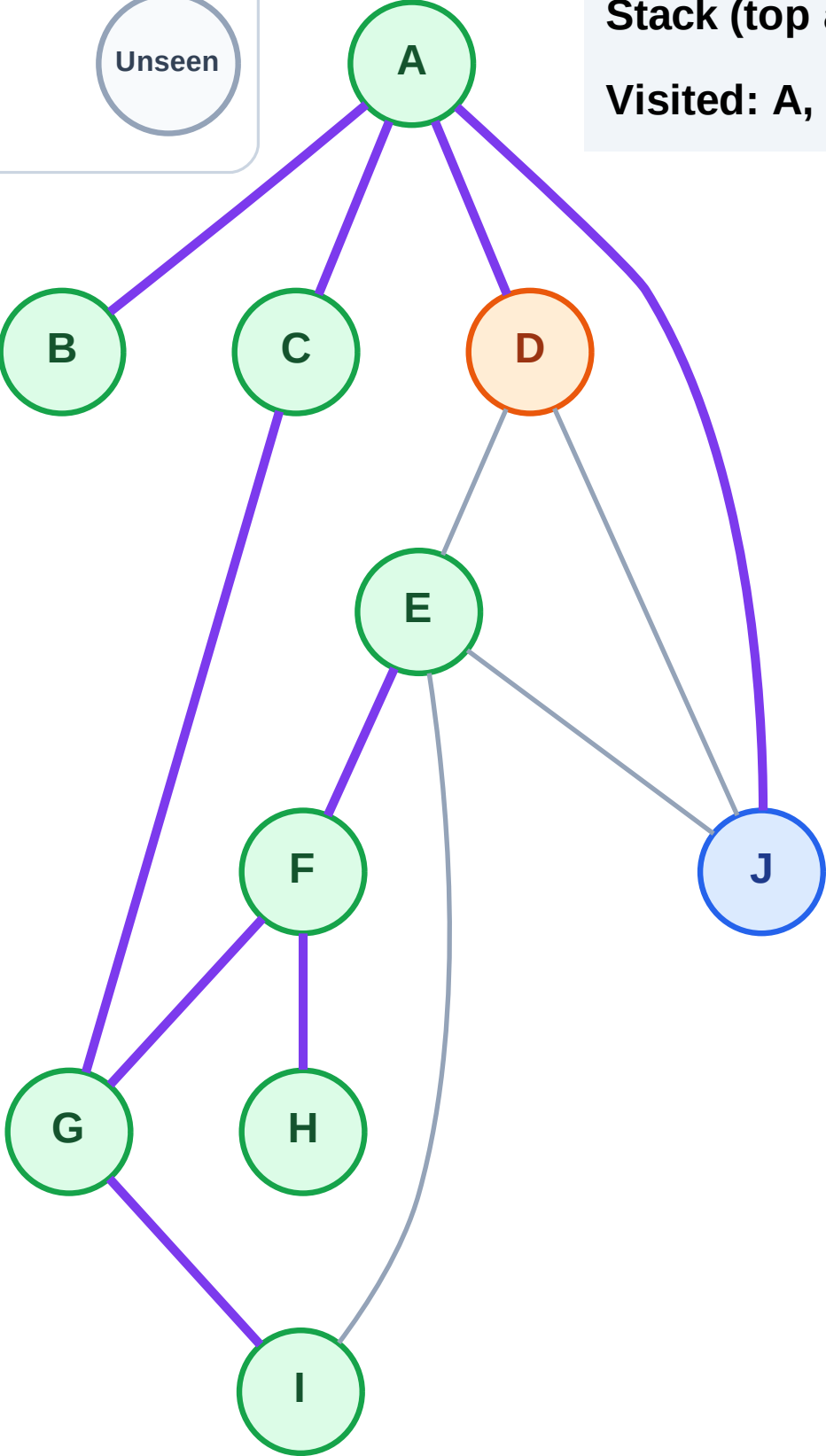
Complete

Processing

Discovered

Unseen

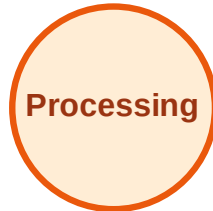
Stack (top at right): J
Visited: A, B, C, E, F, G, H, I



DFS Traversal

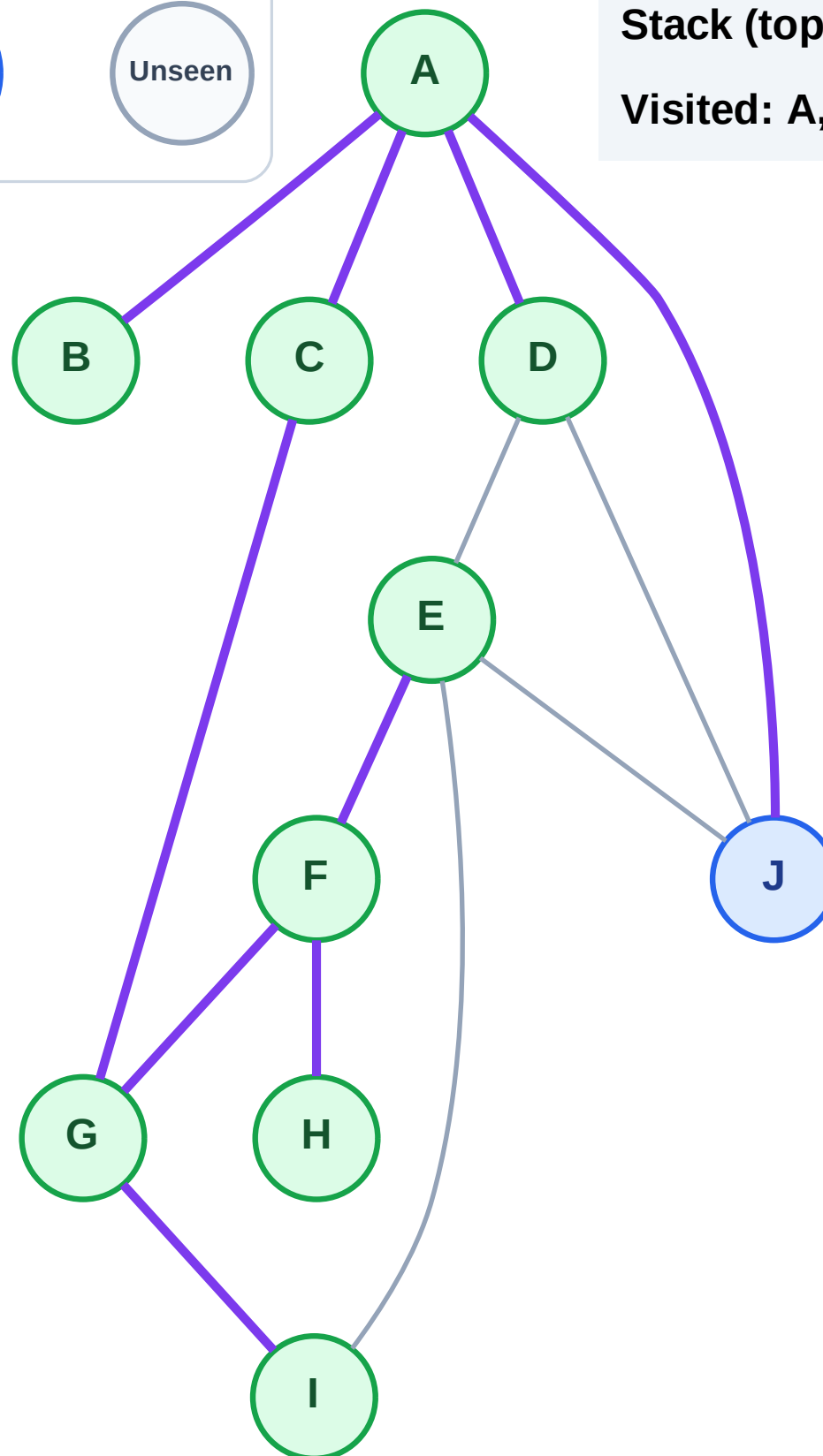
Step 19: No new neighbors from D

Legend



Stack (top at right): J

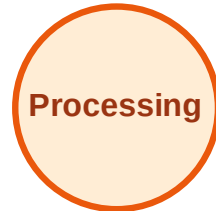
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

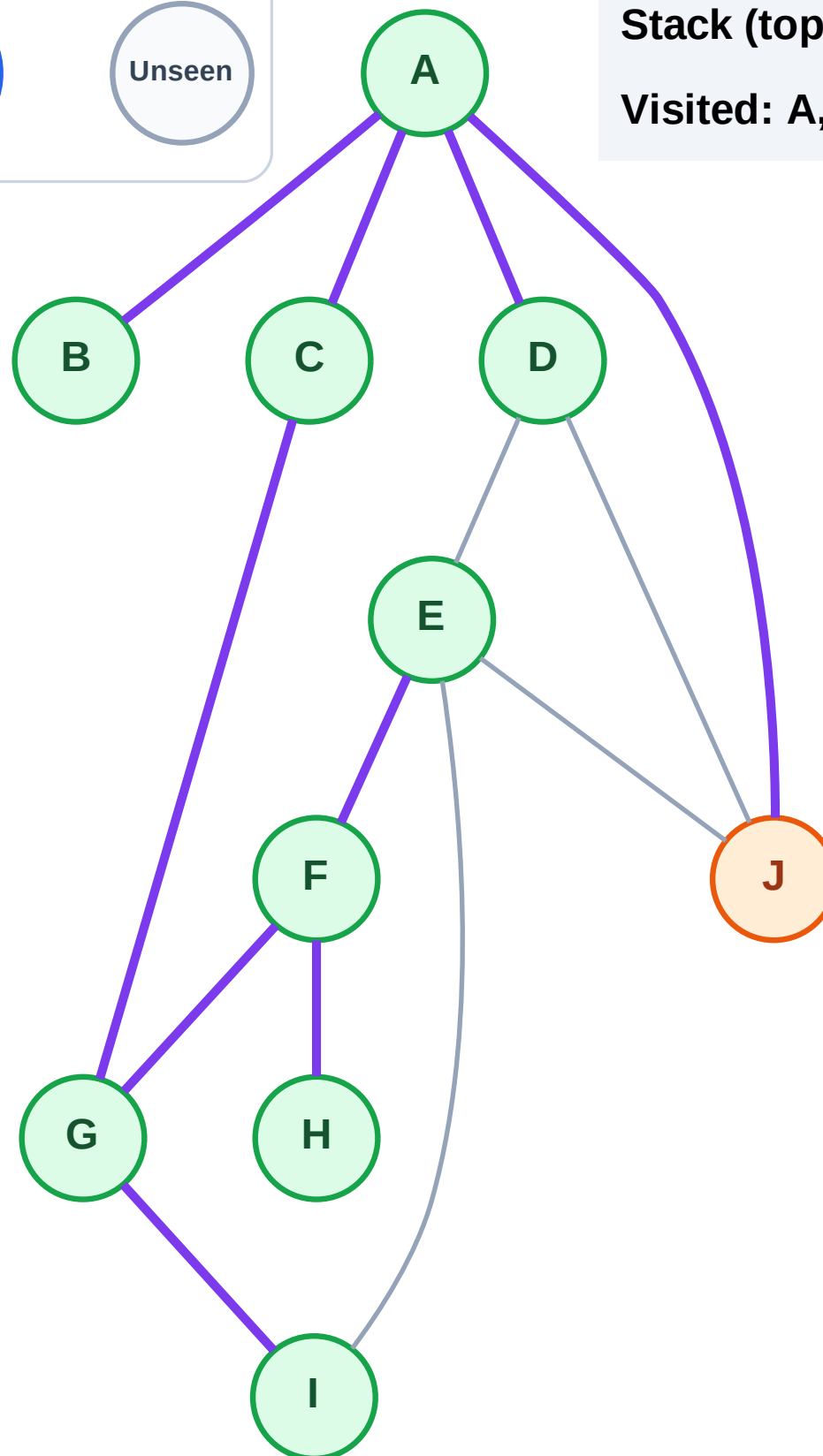
Step 20: Process node J

Legend



Stack (top at right): (empty)

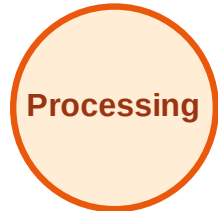
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

